ROBUST VARIATIONAL INFERENCE VIA IMPRECISE PROBABILITIES

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Student id: 11433615

Department of Computer Science

Contents

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List of Tables

List of Figures

Abstract

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Bayesian inference requires the specification of a prior distribution, yet in practice multiple priors are often equally defensible and the resulting posteriors can differ substantially under limited data. This thesis develops a principled framework for robust Bayesian inference under prior uncertainty, grounded in the theory of imprecise probabilities and realised computationally through flow matching. The central theoretical contribution is a guarantee that a credal mixture over K flow-based posterior approximations achieves predictive performance no worse than the best single-prior component: the convexity of KL divergence ensures that combination over a credal set can only improve over selection. An approximate version of this guarantee is established for the practical setting where posteriors are learned by a shared-backbone conditional flow matching model and mixture weights are selected by held-out negative log-likelihood optimisation on the probability simplex.

The framework is evaluated through a systematic experimental programme spanning two real-world Bayesian classification datasets, a regression dataset, and three synthetic settings, with eight weight optimisation algorithms compared across multiple random seeds. Three pre-registered hypotheses are confirmed: the convexity guarantee is empirically visible in the prior-sensitive low-data regime, disappears correctly when data is sufficient to overwhelm prior influence, and the EM algorithm achieves the lowest worst-case regret among all algorithms tested. The credal mixture achieves near-oracle robustness with a factor of eight improvement in worst-case regret over the best val-selected single-prior strategy. A shared-backbone architecture is shown to outperform independently trained flows on predictive performance, training stability, and parameter efficiency, providing empirical validation for the uniform transfer assumption underpinning the theoretical guarantee.

Declaration

No portion of the work referred to in this dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Chapter 1

Introduction

1.1 Background and Motivation

Bayesian inference provides a rigorous framework for updating beliefs about model parameters in the light of observed data, yet the uncertainty it quantifies is not immune to uncertainty itself. This is the problem of second-order uncertainty, which arises because Bayesian inference carries a foundational assumption that the prior distribution is known and correctly specified. When multiple priors are equally plausible, committing to a single one is epistemically insufficient and can lead to overconfident inferences. The framework is precise about uncertainty over parameters but it fails to capture the uncertainty about the choice of the prior which precedes inference at the level of modelling itself. This problem has been recognised in the philosophy of probability since at least the 1960s, formalised under the study of imprecise probabilities and credal sets. Imprecise probability theory replaces the single prior distribution with a credal set, a convex set of distributions representing the range of plausible beliefs, and asks what conclusions are robust across the entire set rather than what is optimal under a single prior (Walley, 1991)(Wheeler, 2023). This is a philosophically distinct response to the problem of second-order uncertainty. Rather than resolving prior uncertainty by collapsing it to a point, the framework carries that uncertainty forward explicitly into the inferential conclusions. The computational challenges of working with sets of distributions are formidable, however, and the focus on scalability in modern machine learning has led to the problem of second-order uncertainty being largely neglected.

Prior sensitivity is a routine challenge in Bayesian inference. When data is scarce, the likelihood is not sufficiently informative to overwhelm the influence of the prior and the posterior remains sensitive to the initial modelling choices. In settings such as healthcare, ecological modelling, or scientific parameter estimation, data is expensive to obtain or inherently limited; two practitioners using equally defensible priors can arrive at substantially different posteriors and therefore different conclusions. In a clinical trial setting, for instance, a sceptical prior and an optimistic prior over treatment efficacy can produce posteriors that disagree substantially about whether a drug should be recommended, not because the data is ambiguous, but because the prior is doing significant inferential work that is never formally acknowledged (Berger, 1994). This sensitivity is not confined to small data or traditional statistical settings. Recent work has shown that prior choice remains consequential in deep Bayesian models; standard priors used in Bayesian neural networks are poorly calibrated and performance is sensitive to prior specification even at scale (Fortuin et al., 2022), and cold posteriors, which correspond to implicitly rescaled priors, have been shown to outperform standard posteriors in deep learning settings (Wenzel et al., 2020). Prior uncertainty is therefore not a historical concern that modern methods have resolved but an active and open problem in probabilistic machine learning.

Variational Inference (VI) has emerged as a powerful tool for Bayesian computation, offering a scalable alternative to Markov Chain Monte Carlo (MCMC) methods. However, standard VI inherits the same foundational limitation: it approximates a single posterior under a fixed prior, leaving second-order uncertainty unaddressed. VI is nevertheless a natural framework for operationalising credal sets at scale: rather than running a separate inference procedure for each candidate prior independently, one can train a family of variational posteriors jointly and combine them into a credal mixture, directly representing the uncertainty that a single posterior cannot capture. The computational challenge is making this tractable; training expressive posterior approximations across a set of priors simultaneously demands a flexible and scalable variational family.

Recent advances in normalising flows and their continuous-time generalisation through neural ordinary differential equations (Chen et al., 2018) have made expressive posterior approximation tractable. Flow matching (Lipman et al., 2023) represents a further step, replacing the computationally expensive log-determinant evaluations required by classical normalising flows with a direct regression objective over a learned velocity field. Crucially, the velocity field can be conditioned on auxiliary information, in this case a discrete prior index, at minimal additional cost, allowing a single shared model to approximate posteriors across all candidate priors simultaneously. This makes flow matching particularly well suited to the credal mixture setting and addresses the computational challenge directly.

This thesis investigates whether credal mixtures of flow-based posteriors can provide robust inference under prior uncertainty. The approach is grounded in a classical result from information theory: the KL divergence is convex in its second argument, which guarantees in principle that a mixture over a set of posteriors performs at least as well as the best individual component (Cover and Thomas, 2006). Whether this guarantee remains empirically visible under deep variational approximation, and what architectural and methodological conditions are required for it to hold, is the central question this work addresses.

1.2 Existing Approaches to Prior Robustness

Several approaches have been proposed to address the problem of prior sensitivity in Bayesian inference and to operationalise the concept of robustness to prior uncertainty. Each offers a different perspective on the problem and a different set of trade-offs, but none provides a principled treatment of second-order uncertainty in the sense formalised by imprecise probability theory.

1.2.1 Epsilon-Contamination and Prior Classes

The original formal treatment of prior robustness models the prior as belonging to an epsilon-contamination class, a convex set of distributions formed by mixing a baseline prior with an arbitrary contaminating distribution within some neighbourhood (Berger, 1984; Insua and Ruggeri, 2000). Posterior bounds are then derived that hold for any prior in the class, providing a formal framework for analysing the sensitivity of inferences to perturbations of the prior. This is the intellectual precursor to the credal set approach and establishes the core idea that working with a class of priors rather than a single one is both principled and tractable. The limitation is that the prior class is defined analytically and the resulting posterior bounds are derived through closed-form calculations, which restricts applicability to simple models and low-dimensional settings. This approach does not scale to modern machine learning problems with high-dimensional parameter spaces and complex likelihoods.

1.2.2 Gamma-Minimax Decision Theory

Rather than committing to one prior, the gamma-minimax framework minimises the maximum expected loss over a class of priors, producing decisions that are robust to the worst-case prior within the class (Berger, 1985). This is philosophically aligned with the imprecise probability approach in that it acknowledges a set of plausible priors rather than a single one. However, the minimax framing collapses the set at decision time: the output is a single decision that hedges against the worst case, rather than a set of posteriors that carries prior uncertainty forward. The second-order uncertainty is acknowledged in the decision procedure but is not propagated through inference in a way that allows downstream conclusions to reflect it.

1.2.3 Bayesian Model Averaging

Bayesian model averaging (BMA) combines a discrete set of candidate models or priors by weighting each according to its marginal likelihood (Hoeting et al., 1999). This is the closest classical precursor to the credal mixture approach and is important to engage with carefully. The key distinction is that BMA uses marginal likelihood weights, which are themselves computed under each model's prior and collapsed into a single weighted predictive distribution. The weights are point estimates and the resulting average is a single distribution rather than a credal set. There is no formal commitment to representing the uncertainty across the set; BMA resolves prior uncertainty by weighting rather than carrying it forward. Furthermore, marginal likelihood weights can be dominated by a single model when one prior fits the data substantially better, which may not reflect genuine robustness across the set.

1.2.4 Stacking of Predictive Distributions

Stacking fits mixture weights over a set of candidate posteriors by maximising held-out log predictive scores, making it the most directly related existing method to the approach developed in this thesis (Yao et al., 2018). Operationally, stacking shares the combination mechanism: a set of posteriors is trained and weights are selected using a likelihood-consistent objective. However, stacking treats the candidate models as fixed and selects weights purely for predictive performance, without any theoretical grounding in imprecise probability theory or the KL convexity guarantee. It is a pragmatic ensemble method rather than a principled treatment of second-order uncertainty. The mixture weights in stacking are justified by cross-validated predictive accuracy alone;

in the credal mixture framework developed here, the combination is grounded in the convexity of KL divergence, which provides a theoretical guarantee that the mixture performs at least as well as the best individual component.

1.2.5 Generalised Bayesian Inference

Generalised Bayesian inference replaces the likelihood in Bayes' rule with a general loss function, producing a family of posteriors indexed by the loss rather than a single posterior (Bissiri et al., 2016). Knoblauch et al. (Knoblauch et al., 2022) extend this framework and explicitly discuss robustness to both prior and likelihood misspecification as a unified problem, making this the most theoretically sophisticated modern approach to Bayesian robustness. The framework implicitly defines a set of plausible posteriors under different modelling assumptions, which is conceptually adjacent to the credal set idea. However, it does not formally invoke imprecise probability theory; it does not treat the resulting set of posteriors as a credal set or ask what can be concluded robustly across it. It produces alternative posteriors rather than representing and propagating genuine second-order uncertainty, and it does not provide a mechanism for combining posteriors across a set of priors in a principled way.

1.2.6 Robust Divergence-Based Variational Inference

Robust divergence-based variational inference replaces the KL divergence in the VI objective with a more robust alternative such as the beta-divergence or gamma-divergence, achieving resistance to outliers and likelihood misspecification (Futami et al., 2018; Knoblauch et al., 2019). While superficially related to the goal of robust inference, this approach addresses a fundamentally different problem. It produces a single posterior under a single prior, changing the objective function to be less sensitive to data contamination but leaving the question of which prior to use entirely unaddressed. A practitioner using these methods still commits to one prior at the outset and obtains one posterior at the end; the second-order uncertainty identified in Section 1.1 is not represented, acknowledged, or propagated.

None of these approaches simultaneously satisfies three properties that the present work aims to achieve: formal grounding in imprecise probability theory, expressive deep variational posterior approximation, and likelihood-consistent combination of posteriors under a credal set of priors. The closest in terms of combination mechanism is stacking, which lacks the theoretical grounding. The closest theoretically is generalised Bayesian inference, which lacks both the credal set framing and the combination mechanism. Imprecise probability theory is philosophically distinct from all of the above: by treating the prior as a credal set from the outset, it propagates uncertainty at the modelling level through to the posterior, asking what can be concluded robustly across all plausible priors rather than optimising under any single one (Walley, 1991; Wheeler, 2023). This thesis operationalises that commitment computationally for the first time in the setting of flow-based variational inference.

1.3 Problem Statement

Having established that existing approaches fall short of a principled treatment of second-order uncertainty, we can now state the specific problem this thesis investigates more precisely. The central challenge is the gap between the theoretical promise of credal mixture inference and its empirical behaviour under deep variational approximation. While the convexity of the KL divergence guarantees that a mixture over a set of posteriors should perform at least as well as the best individual component, it is not clear whether this guarantee remains empirically visible when using deep neural networks to approximate the posteriors. These approximations are learned under a finite data regime and the mixture weights must be estimated from held-out data. These sources of approximation and estimation error may undermine the theoretical guarantee, and it is an open question under what conditions the robust performance of credal mixtures can be realised in practice.

More concretely, the problem has two interacting components. The first is architectural: if each posterior approximation is trained independently under a different prior, there is no guarantee that the resulting approximations are comparable in quality or that their mixture is well-defined as a density. The second is statistical: even with a coherent family of approximations, estimating mixture weights using proxy objectives such as maximum mean discrepancy or adversarial divergence bounds introduces estimation noise that can cause the mixture to underperform its best component. Both failure modes were observed in preliminary experiments and motivate the specific design choices investigated in this thesis.

1.4 Research Objectives

This thesis pursues four concrete objectives in response to the challenges identified above.

1. **Shared-backbone architecture.** Design a flow matching architecture conditioned on a discrete prior index, ensuring that all posterior approximations are trained within a common optimisation framework with comparable capacity across priors.

2. **Likelihood-consistent weight selection.** Investigate held-out negative log likelihood as a principled alternative to proxy divergence objectives such as maximum mean discrepancy and adversarial KL bounds, which were found in preliminary experiments to introduce estimation noise sufficient to mask the theoretical guarantee. Unlike proxy objectives, held-out NLL directly approximates the KL objective underpinning the convexity guarantee, ensuring alignment between the selection criterion and the theoretical motivation.

3. **Empirical evaluation of the convexity guarantee.** Evaluate whether the resulting credal mixtures recover the KL convexity guarantee empirically, using paired per-seed comparisons to control for estimation variance across random initialisations.

4. **Characterisation of failure conditions.** Identify the architectural and statistical conditions under which the theoretical guarantee is and is not empirically visible, contributing to a broader understanding of when classical information-theoretic arguments remain reliable under deep variational approximation.

Taken together, these objectives aim to bridge the gap between the theoretical commitments of imprecise probability theory and their empirical behaviour under deep variational approximation, contributing to a broader understanding of when information-theoretic guarantees survive the transition from exact to approximate inference.

1.5 Report Structure

The remainder of this thesis is organised as follows.

Chapter 2 presents the theoretical foundations underpinning the approach. It introduces Bayesian inference and variational inference, formalises imprecise probability theory and credal sets, and establishes the KL convexity guarantee that motivates the credal mixture framework. It then develops the normalising flow and flow matching machinery, and proves the approximate mixture bound that characterises the conditions under which the guarantee survives flow-based approximation and finite held-out weight selection.

Chapter 3 describes the methodology. It formalises the three research hypotheses and the two-regime experimental design, introduces the datasets and credal sets used for each problem, and describes the shared-backbone `VelocityResNet` architecture and its prior conditioning mechanism. The two-phase training procedure, the eight credal weight optimisation algorithms compared, and the full evaluation protocol are also defined here.

Chapter 4 presents the results and discussion. The primary evaluation on German Credit in the low-data regime confirms H1 and H3, demonstrating that the credal mixture achieves near-oracle worst-case regret and that EM is the most consistent weight optimisation algorithm. The full-data null control confirms H2. An architectural ablation validates the shared-backbone design, and supplementary analyses examine prior sensitivity, weight stability, leave-one-prior-out behaviour, calibration, and optimiser convergence.Section 4.5.6 presents a speed comparison between the flow matching training objective and a CNF baseline using the same architecture, validating the computational motivation of Section 2.5.

Chapter 5 concludes the thesis with a summary of contributions, an acknowledgement of limitations, and directions for future work including continuous credal sets, theoretical bounds on the approximation gap, and extension to Bayesian neural networks.

Chapter 2

Literature Review and Theory

2.1 Bayesian Inference

Bayesian inference provides a rational framework for reasoning under uncertainty, grounded in the principle that a rational agent should update their beliefs in accordance with Bayes' rule upon observing new evidence (Bernardo and Smith, 1994). In this section we formally introduce the posterior distribution and the marginal likelihood, and discuss the computational intractability that motivates the approximate inference methods developed in subsequent sections.

2.1.1 Probability Notation and Conventions

We work within the standard measure-theoretic framework of probability theory. A probability space is $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a sample space, $\mathcal{F}$ is a σ-algebra of measurable events, and $\mathbb{P} : \mathcal{F} \to [0,1]$ is a probability measure satisfying $\mathbb{P}(\Omega) = 1$. A random variable $X : \Omega \to \mathcal{X}$ has distribution $P = \mathbb{P} \circ X^{-1}$; when P is absolutely continuous with respect to a reference measure μ we write its density as $p = dP/d\mu$ and work with densities throughout, suppressing explicit reference to μ where context is clear.

Probability density functions and probability mass functions are both denoted $p(\cdot)$, with the distinction determined by context. We use the terms *distribution* and *density* interchangeably. Conditional densities are written $p(\cdot \mid \cdot)$ following the convention of Gelman et al. Gelman et al. (2013). As an exception to the general $p(\cdot)$ convention, we reserve $\pi(\theta)$ specifically for prior distributions over parameters. This distinction becomes important in later sections where multiple priors $\pi_1, \ldots, \pi_K$ are considered simultaneously. The following symbols are used consistently throughout this thesis.

- $(\Omega, \mathcal{F}, \mathbb{P})$: underlying probability space.

- $\theta \in \Theta$: model parameters on the parameter space Θ.
- $x \in \mathcal{X}$: observed data on the observation space $\mathcal{X}$.
- $\pi(\theta)$: prior distribution over parameters, reserved exclusively for priors throughout this thesis.
- $\mathcal{L}(x \mid \theta)$: likelihood function.
- $p(\theta \mid x)$: posterior distribution.
- $Z = \int \mathcal{L}(x \mid \theta)\,\pi(\theta)\,d\theta$: normalising constant.

Further notation is introduced as needed when each concept is formally defined.

2.1.2 The Posterior and Bayes' Rule

Let $\theta \in \Theta$ denote the unknown parameters and $x \in \mathcal{X}$ the observed data. The joint distribution factorises as

$$p(\theta, x) = \mathcal{L}(x \mid \theta)\,\pi(\theta) \tag{2.1}$$

and conditioning on the observed data yields the posterior:

$$p(\theta \mid x) = \frac{\mathcal{L}(x \mid \theta)\,\pi(\theta)}{\int_\Theta \mathcal{L}(x \mid \theta)\,\pi(\theta)\,d\theta} \tag{2.2}$$

The denominator $p(x)$, known as the marginal likelihood or evidence, serves as a normalising constant ensuring the posterior is a valid probability distribution.

2.1.3 Posterior Intractability

The posterior $p(\theta \mid x)$ is the central object of Bayesian inference, yet computing it exactly is intractable for most models of practical interest. The difficulty lies entirely in the normalising constant Z. The numerator $\mathcal{L}(x \mid \theta)\,\pi(\theta)$ is straightforward to evaluate pointwise for any given θ, since it requires only evaluating the likelihood and the prior. The denominator, however, requires integrating this product over the entire parameter space:

$$Z = \int_\Theta \mathcal{L}(x \mid \theta)\,\pi(\theta)\,d\theta \tag{2.3}$$

This intractability has two important consequences. First, we cannot evaluate the posterior density exactly at any point θ, since doing so requires knowing Z. Second, we cannot draw exact samples from the posterior, which would otherwise allow us to approximate posterior expectations via Monte Carlo methods. Both limitations mean that exact Bayesian inference is computationally infeasible in practice, and approximate inference methods are necessary.

This intractability is particularly consequential in the context of this thesis. When multiple priors $\pi_1, \ldots, \pi_K$ are under consideration, we face not one but K intractable posteriors simultaneously. Each posterior $p(\theta \mid x, \pi_k)$ has its own normalising constant Z_k, and the difficulty of computing any one of them is compounded by the need to reason about the entire family. This motivates approximate inference methods that are both scalable and amenable to conditioning on auxiliary information such as the prior index k, a requirement that guides the architectural choices developed in subsequent sections.

Among the available approaches: Laplace approximations (MacKay, 1992), expectation propagation (Minka, 2001), integrated nested Laplace approximations (Rue et al., 2009), and approximate Bayesian computation (Sisson et al., 2018) and we focus on variational inference, which recasts posterior inference as an optimisation problem, making it scalable, differentiable, and amenable to conditioning on auxiliary information (Blei et al., 2017).

2.2 Variational Inference

The key insight behind variational inference is that we can turn the problem of inference into a problem of optimisation (Blei et al., 2017). Instead of computing $p(\theta \mid x)$ exactly, we pick a family of tractable distributions $\mathcal{Q}$ and find the member closest to the true posterior, transforming an intractable integration problem into one we can solve with gradient-based optimisation.

2.2.1 The Variational Objective

Let $q(\theta) \in \mathcal{Q}$ be a tractable approximation to $p(\theta \mid x)$. The standard variational objective minimises the reverse KL divergence:

$$q^* = \arg\min_{q \in \mathcal{Q}} \mathrm{KL}(q \| p(\cdot \mid x)) \tag{2.4}$$

Expanding the KL divergence reveals that minimising it requires evaluating $\log p(\theta \mid x) = \log \mathcal{L}(x \mid \theta) + \log \pi(\theta) - \log Z$, which involves the intractable $\log Z$. This motivates the tractable surrogate derived in the following section.

Remark 2.1. The KL divergence $\mathrm{KL}(q \| p)$ is asymmetric: the reverse direction $\mathrm{KL}(q \| p)$, used here, penalises q for placing mass where the posterior has none, while the forward direction $\mathrm{KL}(p\|q)$ would require integrating against the intractable posterior and is not directly usable as a variational objective. However, the forward KL plays a central role in Section 2.4, where it appears in the theoretical guarantees for credal mixture inference.

2.2.2 The Evidence Lower Bound

Since $\mathrm{KL}(q\|p)$ cannot be minimised directly due to the intractability of $\log Z$, we derive a tractable surrogate. Starting from the log marginal likelihood and multiplying by $1 = \int q(\theta)\,d\theta$:

$$\begin{aligned}\log p(x) &= \int q(\theta)\log p(x)\,d\theta \\ &= \int q(\theta)\log\frac{p(\theta,x)}{p(\theta\mid x)}\,d\theta \\ &= \int q(\theta)\log\frac{p(\theta,x)}{q(\theta)}\,d\theta + \int q(\theta)\log\frac{q(\theta)}{p(\theta\mid x)}\,d\theta \\ &= \mathcal{L}(q) + \mathrm{KL}(q\|p) \end{aligned} \tag{2.5}$$

where the Evidence Lower Bound (ELBO) is defined as:

$$\mathcal{L}(q) = \int q(\theta)\log\frac{p(\theta,x)}{q(\theta)}\,d\theta = \mathbb{E}_q\left[\log p(\theta,x)\right] - \mathbb{E}_q\left[\log q(\theta)\right] \tag{2.6}$$

Since $\mathrm{KL}(q\|p) \geq 0$ with equality if and only if $q = p(\cdot\mid x)$, it follows that $\log p(x) \geq \mathcal{L}(q)$, explaining the name. Since $\log p(x)$ is constant with respect to q, maximising the ELBO is equivalent to minimising the KL divergence:

$$\operatorname*{arg\,max}_{q\in\mathcal{Q}} \mathcal{L}(q) = \operatorname*{arg\,min}_{q\in\mathcal{Q}} \mathrm{KL}(q\|p) \tag{2.7}$$

The ELBO can be further decomposed by writing $p(\theta,x) = \mathcal{L}(x\mid\theta)\,\pi(\theta)$:

$$\mathcal{L}(q) = \mathbb{E}_q\left[\log\mathcal{L}(x\mid\theta)\right] - \mathrm{KL}(q\|\pi) \tag{2.8}$$

The first term encourages q to place mass on parameter values that explain the data well; the second penalises deviation from the prior, acting as a regulariser. Crucially, this objective depends on the choice of $\pi(\theta)$: different priors produce different optimal approximations q^*, connecting directly to the prior sensitivity problem motivating this thesis.

2.2.3 Mean Field Variational Inference

The simplest and most widely used variational family is the mean field approximation (Opper and Saad, 2001). In this approach the variational distribution is assumed to factorise fully over the components of θ:

$$q(\theta) = \prod_{j=1}^{D} q_j(\theta_j) \tag{2.9}$$

where each q_j is an independent marginal distribution over the j-th component. This factorisation reduces the problem of optimising over a joint distribution on a D-dimensional space to optimising over D independent one-dimensional distributions, which is substantially more tractable.

Under the mean field assumption, the ELBO can be optimised using coordinate ascent variational inference (CAVI), which iteratively updates each factor q_j while holding the others fixed (Bishop, 2006; Blei et al., 2017). The optimal update for factor q_j is:

$$\log q_j^*(\theta_j) = \mathbb{E}_{q_{-j}}[\log p(\theta, x)] + \text{const} \tag{2.10}$$

where $\mathbb{E}_{q_{-j}}$ denotes the expectation with respect to all factors except q_j. This update has a closed form for models in the exponential family, making mean field VI computationally efficient in those settings (Wainwright and Jordan, 2008).

The independence assumption underlying mean field VI is however a strong one. Posterior distributions over model parameters often exhibit complex dependencies that the factorial approximation cannot represent. In a Bayesian linear regression model, for instance, the posterior over weights and noise variance are typically correlated, and forcing independence causes the mean field approximation to misrepresent the true uncertainty structure. Structured mean field approximations relax this by allowing dependencies within groups of variables while maintaining independence between groups (Saul and Jordan, 1996), but remain fundamentally limited by between-group independence assumptions.

2.2.4 Limitations of Standard Variational Families

The mean field approximation and its structured variants share a common weakness: the expressiveness of the variational family $\mathcal{Q}$ fundamentally limits the quality of the approximation. No matter how well the ELBO is optimised, if the true posterior lies outside $\mathcal{Q}$, the best approximation within $\mathcal{Q}$ will remain a poor representation of the true uncertainty. This is known as the approximation gap, distinct from the optimisation gap that arises from imperfect ELBO maximisation (Turner and Sahani, 2011).

Beyond expressiveness, standard variational families have a second limitation that is particularly relevant to this thesis. The ELBO depends explicitly on the prior $\pi(\theta)$ through the regularisation term $\mathrm{KL}(q\|\pi)$, meaning different priors produce different optimal approximations even when the likelihood and data are held fixed:

$$q_k^* = \arg\min_{q\in\mathcal{Q}} \mathrm{KL}(q\|p(\cdot \mid x, \pi_k)), \quad k = 1, \dots, K \tag{2.11}$$

Standard variational inference provides no mechanism for reasoning about this family of approximations jointly. Each prior produces one approximate posterior, and the framework offers no principled way to combine them or to ask what can be concluded robustly across the set. This is precisely the problem of second-order uncertainty identified in Chapter 1, now stated in the language of variational inference.

Addressing this limitation requires two things. First, a variational family expressive enough to approximate complex posteriors faithfully, so that the approximation gap does not obscure differences between posteriors under different priors. Second, a principled mechanism for combining the K approximate posteriors. The theoretical justification for the second requirement is provided by the KL convexity guarantee of Section 2.4. The first motivates the use of normalising flows and flow matching as the variational family, introduced in Sections 2.5 and 2.6.

2.3 Imprecise Probabilities and Credal Sets

As established in Section 2.1, Bayesian inference requires a prior $\pi(\theta)$. When multiple priors are equally consistent with prior knowledge, committing to a single one introduces unjustified precision, and the resulting posterior inherits that commitment silently. Imprecise probability theory addresses this by replacing the single prior with a set of distributions, representing the degree of modelling uncertainty explicitly rather than suppressing it.

2.3.1 Motivation: When Precise Priors Are Unjustified

Knight (1921) drew a fundamental distinction between risk, where outcomes can be assigned precise probabilities, and uncertainty, where the evidence is insufficient to justify any single probability assignment. Kass and Greenhouse (1989) demonstrate this concretely: in a clinical meta-analysis, three defensible priors applied to the same dataset produce qualitatively different conclusions about treatment effectiveness, with one prior yielding a significant treatment effect and another yielding none. The data does not resolve the disagreement because the posterior remains sensitive to the prior in the limited data regime. Wenzel et al. (2020) show the same phenomenon persists in Bayesian neural networks, where prior choice produces differences of several percentage points in predictive accuracy even under large training sets.

These examples share a common structure: multiple priors are defensible, each produces a materially different posterior, and standard Bayesian inference provides no mechanism for acknowledging or propagating this disagreement.

2.3.2 Lower and Upper Previsions

The foundational response is to replace a single distribution with a set $\mathcal{P}$ consistent with the available evidence, working with lower and upper probabilities:

$$\underline{P}(A) = \inf_{P \in \mathcal{P}} P(A), \qquad \overline{P}(A) = \sup_{P \in \mathcal{P}} P(A) \tag{2.12}$$

The gap $\overline{P}(A) - \underline{P}(A)$ quantifies irreducible imprecision. The framework generalises to arbitrary bounded functions f via lower and upper previsions (Walley, 1991):

$$\underline{P}(f) = \inf_{P \in \mathcal{P}} \mathbb{E}_P[f], \qquad \overline{P}(f) = \sup_{P \in \mathcal{P}} \mathbb{E}_P[f] \tag{2.13}$$

These reduce to ordinary expectations when $\mathcal{P}$ is a singleton, recovering standard probability theory. The key coherence property is:

$$\overline{P}(f) = -\underline{P}(-f) \tag{2.14}$$

A lower prevision satisfying Walley's coherence axioms is called a coherent lower prevision, ensuring the framework does not lead to sure loss.

2.3.3 Credal Sets: Formal Definition and Properties

The set of probability distributions $\mathcal{P}$ underlying the lower and upper previsions is called a credal set. We now define this object formally.

Definition 2.2 (Credal Set)**.** A credal set $\mathcal{P}$ is a closed convex set of probability distributions over a measurable space $(\Omega, \mathcal{F})$. Each element $P \in \mathcal{P}$ is a probability distribution consistent with the available evidence and prior knowledge.

The convexity requirement is natural: if P_1 and P_2 are both consistent with the available evidence, any mixture $\lambda P_1 + (1-\lambda)P_2$ for $\lambda \in [0,1]$ should also be consistent, since it represents a further hedging between two defensible positions. The closure requirement ensures that the infimum and supremum in the prevision definitions are attained (Augustin et al., 2014). A precise distribution is the degenerate case $\mathcal{P} = \{P\}$; a larger credal set corresponds to greater modelling uncertainty.

Several standard constructions give rise to credal sets in practice. The epsilon-contamination class (Berger, 1984):

$$\mathcal{P}_\varepsilon = \left\{(1-\varepsilon)P_0 + \varepsilon Q : Q \in \mathcal{M}\right\} \tag{2.15}$$

models the prior as a mixture of a baseline distribution P_0 and an arbitrary contaminating distribution Q, with ε controlling the degree of imprecision. This construction is the classical approach to prior robustness and the intellectual precursor to the credal set framework used in this thesis.

In this thesis we work with a finite credal set of K priors $\Pi = \{\pi_1, \ldots, \pi_K\}$, each representing a plausible prior that cannot be ruled out given the available knowledge. The convex hull $\mathrm{conv}(\Pi) = \{\sum_{k=1}^{K} w_k \pi_k : w \in \Delta_{K-1}\}$ is itself a credal set, and working with the finite generating set Π rather than the full convex hull is a common and computationally convenient simplification (Cozman, 2000).

Applying Bayes' rule to each element of Π induces a set of posteriors $Q^* = \{p(\theta \mid x, \pi_k) : k = 1, \ldots, K\}$. Rather than reporting this full set, which is not directly tractable, we form the credal mixture:

$$q_w(\theta) = \sum_{k=1}^{K} w_k q_k^\theta(\theta), \quad w \in \Delta_{K-1} \tag{2.16}$$

where each q_k^θ is a flow-based approximation to $p(\theta \mid x, \pi_k)$. This is a single tractable distribution that represents the credal set without collapsing it to a point estimate under any single prior. The notation for exact and approximate posteriors is formalised in Definition 2.3 in Section 2.4, where it is first required.

2.4 The KL Convexity Guarantee

The credal mixture $q_w = \sum_{k=1}^{K} w_k q_k^\theta$ is a tractable representation of the credal set, but its theoretical justification requires answering whether combining posteriors under different priors produces a better approximation than simply choosing the best single one. In this section we show that the answer is yes, via a classical property of KL divergence. We first fix notation for the exact and approximate posteriors that appear throughout.

Definition 2.3 (Exact and Approximate Posteriors)**.** For each prior $\pi_k \in \Pi$, the exact posterior is $q_k^*(\theta) = p(\theta \mid x, \pi_k)$ and the flow-based approximation is q_k^θ, obtained by training the velocity field $v_\theta(z,t,k)$ with the objective of Section 2.6. The approximation error of the k-th component is $\varepsilon_k = \mathrm{KL}(q_k^* \| q_k^\theta)$.

2.4.1 Convexity of KL Divergence

Lemma 2.4 (Convexity of KL in its Second Argument)**.** *For any fixed distribution P and any $\lambda \in [0,1]$:*

$$\mathrm{KL}(P \| \lambda Q_1 + (1-\lambda) Q_2) \leq \lambda \, \mathrm{KL}(P \| Q_1) + (1-\lambda) \, \mathrm{KL}(P \| Q_2) \tag{2.17}$$

Proof. Since $-\log$ is convex, Jensen's inequality gives $-\log(\lambda a + (1-\lambda) b) \leq -\lambda \log a - (1-\lambda) \log b$. Setting $a = q_1(\theta)/p(\theta)$, $b = q_2(\theta)/p(\theta)$, rearranging, multiplying by $p(\theta) \geq 0$, and integrating over Θ yields the result. □

Remark 2.5. Lemma 2.4 extends to K components by induction: for any $w \in \Delta_{K-1}$, $\mathrm{KL}(P \| \sum_{k=1}^{K} w_k Q_k) \leq \sum_{k=1}^{K} w_k \, \mathrm{KL}(P \| Q_k)$.

2.4.2 The Mixture Lower Bound

Theorem 2.6 (Exact Mixture Lower Bound)**.** *Let $p(\theta \mid x)$ be the true posterior and $q_1, \ldots, q_K$ any K distributions on Θ. Then:*

$$\inf_{w \in \Delta_{K-1}} \mathrm{KL}(p(\theta \mid x) \| q_w) \leq \min_{k \in \{1,\ldots,K\}} \mathrm{KL}(p(\theta \mid x) \| q_k) \tag{2.18}$$

Proof. Let $k^* \in \arg\min_k \mathrm{KL}(p(\theta \mid x) \| q_k)$ and define $e_{k^*} \in \Delta_{K-1}$ with $(e_{k^*})_k = \mathbf{1}[k = k^*]$. Since e_{k^*} is feasible:

$$\inf_{w \in \Delta_{K-1}} \mathrm{KL}(p(\theta \mid x) \| q_w) \leq \mathrm{KL}(p(\theta \mid x) \| q_{e_{k^*}}) = \min_k \mathrm{KL}(p(\theta \mid x) \| q_k) \tag{2.19}$$

□

2.4.3 Implications for Credal Mixture Inference

Theorem 2.6 establishes that the optimal mixture over the credal set performs at least as well as the best single component under KL divergence from the true posterior. Three implications follow directly for the credal mixture framework developed in this thesis.

First, it provides theoretical justification for combining posteriors across the credal set rather than selecting one. No matter how good the best single posterior is, the optimal mixture is at least as good. The combination can only help, never hurt, in the exact setting.

Second, it establishes that the simplex Δ_{K-1} is the right domain for the weight optimisation problem. The degenerate weight vectors at the vertices of the simplex correspond to the single-component posteriors, and the interior of the simplex corresponds to genuine mixtures. The theorem guarantees that the optimum lies somewhere in the simplex and is at least as good as any vertex.

Third, and most importantly for this thesis, the result motivates the choice of held-out negative log-likelihood as the weight selection objective. Since the theorem is stated in terms of KL divergence from the true posterior, and minimising KL divergence from the true data generating distribution is equivalent to maximising the expected log likelihood, the held-out NLL is the natural empirical proxy for the theoretical objective. This connection is formalised in Section 2.7.

Remark 2.7. Theorem 2.6 is stated for exact posteriors. In practice, the posteriors q_k are flow-based approximations q_k^θ and the weights are estimated from finite held-out data. Whether the guarantee survives these approximations depends on the quality of the variational family and the weight selection procedure, which is the central empirical question of Chapter 4. The theoretical conditions are established in Section 2.7.

2.5 Normalising Flows

Normalising flows construct a complex distribution by starting from a simple base distribution and transforming it through learned invertible mappings. This addresses the core limitation of mean field VI: because the transformations are learned rather than fixed, the resulting distributions can represent arbitrary posterior geometry including strong correlations, multiple modes, and heavy tails (Papamakarios et al., 2021).

Even with expressive flows, the credal mixture framework requires training K separate approximations. When trained independently there is no guarantee they are comparable in quality: one prior may lead to a well-behaved posterior that the flow approximates easily, while another may lead to complex geometry that the same architecture struggles with. Mixing approximations of uneven quality does not reliably recover the convexity guarantee. What is needed is a flow family amenable to conditioning on the prior index k, so a single model can approximate all K posteriors within a shared training framework. The computational bottleneck that motivates the specific approach used here is described in Section 2.5.3.

2.5.1 Change of Variables and the Flow Framework

The key mathematical operation underpinning normalising flows is the change of variables formula. When a random variable is passed through an invertible transformation, its density changes in a predictable way determined by how much the transformation locally stretches or compresses space. Formalising this relationship allows us to construct complex distributions by composing simple, learned transformations, which is precisely what we need for a flexible variational family.

Let $Z \sim p_Z$ on $\mathbb{R}^D$ and let $f : \mathbb{R}^D \to \mathbb{R}^D$ be differentiable and invertible. The density of $X = f(Z)$ is:

$$p_X(x) = p_Z(z) \left| \det \frac{\partial f(z)}{\partial z} \right|^{-1}, \quad z = f^{-1}(x) \tag{2.20}$$

A normalising flow composes L invertible transformations $f_1, \ldots, f_L$ applied to $z_0 \sim p_Z$, giving log density:

$$\log p_X(x) = \log p_Z(z_0) - \sum_{\ell=1}^{L} \log \left| \det \frac{\partial f_\ell(z_{\ell-1})}{\partial z_{\ell-1}} \right| \tag{2.21}$$

In the variational inference setting, p_X plays the role of $q(\theta)$ and the flow parameters are optimised to maximise the ELBO using equation (2.21) (Rezende and Mohamed, 2015). Early architectures such as RealNVP (Dinh et al., 2017) and Glow (Kingma and Dhariwal, 2018) used triangular Jacobians whose determinants cost only $O(D)$ to compute, enabling practical training.

2.5.2 Continuous Normalising Flows and Neural ODEs

The continuous-time limit of discrete flows replaces the finite composition of transformations with a differential equation (Chen et al., 2018). Rather than applying L discrete invertible maps, a velocity field $v_\theta : \mathbb{R}^D \times [0,1] \to \mathbb{R}^D$ governs the continuous evolution of particles:

$$\frac{dz(t)}{dt} = v_\theta(z(t),t), \quad z(0) \sim p_Z \tag{2.22}$$

The solution $z(1)$, obtained by integrating equation (2.22) from $t = 0$ to $t = 1$, defines a sample from the approximate posterior. The log density of a sample evolves via the instantaneous change of variables formula (Chen et al., 2018):

$$\frac{d \log p(z(t))}{dt} = -\mathrm{tr}\left(\frac{\partial v_\theta(z(t),t)}{\partial z(t)}\right) \tag{2.23}$$

where $\mathrm{tr}(\cdot)$ denotes the matrix trace. This is the continuous analogue of the log Jacobian determinant sum in equation (2.21), and it follows from the Liouville equation in fluid dynamics, which relates the rate of change of a probability density to the divergence of the velocity field.

Continuous normalising flows offer two key advantages over discrete flows. First, the architecture of v_θ is unconstrained: unlike coupling layer architectures which impose specific triangular Jacobian structure, any neural network can serve as v_θ, making CNFs strictly more expressive. Second, the trace in equation (2.23) costs $O(D^2)$ exactly and can be approximated in $O(D)$ using the Hutchinson trace estimator (Hutchinson, 1990; Grathwohl et al., 2019), compared to the $O(D^3)$ determinant required by general discrete flows.

Despite these advantages, CNFs introduce a new computational challenge. Density evaluation is now coupled to ODE integration: to evaluate $\log p(z(1))$, one must integrate both the ODE and the trace term in equation (2.23) simultaneously from $t = 0$ to $t = 1$. This coupling between simulation and density evaluation is absent in discrete flows, where the log Jacobian is computed analytically at each transformation layer. The implications of this coupling for the credal mixture setting, and the computational approach that resolves it, are developed in the following section.

2.5.3 The Log-Determinant Bottleneck

Training a CNF requires integrating the ODE in equation (2.22) numerically at every training step to evaluate the log density via equation (2.23). A standard adaptive solver such as Dormand-Prince (Dormand and Prince, 1980) requires on the order of hundreds of function evaluations per integration, making training substantially more expensive than discrete flows. In the credal mixture setting where K posteriors must be approximated, this cost multiplies by K under independent training, making the approach increasingly impractical as the credal set grows.

Even setting aside simulation cost, evaluating the trace in equation (2.23) requires $O(D^2)$ computation exactly, or a stochastic $O(D)$ Hutchinson approximation that introduces variance into the training objective and can destabilise optimisation (Grathwohl et al., 2019). Both bottlenecks share a common structure: they arise because density evaluation is coupled to ODE simulation. An approach that trains the velocity field directly without density evaluation or ODE simulation during training would eliminate both simultaneously. The following section introduces exactly such an approach.

2.6 Flow Matching

Flow matching reframes the training objective entirely. Rather than asking v_θ to reproduce the exact log density at each training step, it asks v_θ to match a known target vector field that generates a desired probability path from source to target. This is a direct regression problem requiring no ODE simulation and no density evaluation during training (Lipman et al., 2023; Albergo and Vanden-Eijnden, 2022).

The key insight is that while the marginal target vector field is intractable, it can be replaced by a conditional target vector field that conditions on individual data points. The two objectives share the same gradient with respect to θ, so training on the tractable conditional objective is theoretically equivalent to training on the intractable marginal one (Lipman et al., 2023).

Beyond training efficiency, the velocity field $v_\theta(z,t)$ can be extended to $v_\theta(z,t,k)$ by conditioning on a discrete prior index k at minimal additional cost. A single shared model then approximates all K posteriors simultaneously, directly addressing the architectural failure mode of independent training and satisfying the uniform approximation error condition of Assumption 2.2.

2.6.1 Probability Paths and Vector Fields

Definition 2.8 (Probability Path)**.** A probability path is a family $\{p_t\}_{t\in[0,1]}$ satisfying $p_0 = p_Z$ and $p_1 \approx p_{\text{target}}$.

A velocity field v and path $\{p_t\}$ are compatible if v generates $\{p_t\}$, i.e., integrating the ODE from $z(0) \sim p_0$ produces $z(t) \sim p_t$ for all t. The condition for compatibility is the continuity equation:

$$\frac{\partial p_t(z)}{\partial t} = -\nabla \cdot (p_t(z)\, v(z,t)) \tag{2.24}$$

The velocity field generating a given path is not unique, but all compatible fields produce the same marginal distributions and the same samples at $t = 1$.

2.6.2 The Conditional Flow Matching Objective

The natural training objective is the flow matching loss:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1],\, z\sim p_t} \|v_\theta(z,t) - u_t(z)\|^2 \tag{2.25}$$

where $u_t(z)$ is the marginal vector field generating p_t. This is intractable because $u_t(z)$ requires integrating over all target points. The resolution is to condition on individual samples $z_1 \sim p_1$ and define conditional paths $p_t(z \mid z_1) = \mathcal{N}(z \mid \mu_t(z_1), \sigma_t(z_1)^2 I)$ with boundary conditions matching p_0 at $t = 0$ and concentrating at z_1 at $t = 1$. The conditional vector field $u_t(z \mid z_1)$ generating this path is available in closed form, giving the conditional flow matching objective:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t,\, z_1\sim p_1,\, z\sim p_t(\cdot|z_1)} \|v_\theta(z,t) - u_t(z \mid z_1)\|^2 \tag{2.26}$$

Theorem 2.9 (Equivalence of Flow Matching Objectives (Lipman et al., 2023))**.** $\nabla_\theta \mathcal{L}_{FM}(\theta) = \nabla_\theta \mathcal{L}_{CFM}(\theta)$.

Proof. Expanding $\mathcal{L}_{\text{FM}}$ and writing $p_t(z) = \mathbb{E}_{z_1}[p_t(z \mid z_1)]$:

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t,z\sim p_t} \|v_\theta\|^2 - 2\,\mathbb{E}_{t,z\sim p_t}\left[v_\theta^\top u_t(z)\right] + C \tag{2.27}$$

The cross term equals $\mathbb{E}_{t,z_1,z\sim p_t(\cdot|z_1)}[v_\theta^\top u_t(z \mid z_1)]$ by the law of total expectation, as does the first term. Combining gives $\mathcal{L}_{\text{FM}}(\theta) = \mathcal{L}_{\text{CFM}}(\theta) + C'$ where C' is constant in θ. □

Theorem 2.9 makes flow matching practical: the conditional objective, which requires no ODE simulation, produces the same velocity field as the intractable marginal objective.

2.6.3 Simulation-Free Training

Theorem 2.9 establishes that the conditional flow matching objective is equivalent to the marginal objective for optimisation purposes. What remains is to specify a concrete probability path and show that the resulting training procedure requires no ODE integration at any point during training.

The simplest and most effective choice is the linear interpolant path, also called the optimal transport path (**?**). For a target sample $z_1 \sim p_1$ and a base sample $z_0 \sim p_Z$, define:

$$z_t = (1-t)z_0 + tz_1, \quad t \in [0,1] \tag{2.28}$$

This is simply a straight line in $\mathbb{R}^D$ connecting z_0 to z_1. The conditional vector field that generates this path is obtained by differentiating with respect to t:

$$u_t(z \mid z_1) = \frac{dz_t}{dt} = z_1 - z_0 \tag{2.29}$$

This is a constant vector field pointing directly from z_0 to z_1, available in closed form with no approximation required. Substituting into the conditional flow matching objective gives the concrete training objective used in this thesis:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t\sim\mathcal{U}[0,1],z_0\sim p_Z,z_1\sim p_1} \|v_\theta(z_t,t) - (z_1 - z_0)\|^2 \tag{2.30}$$

where $z_t = (1-t)z_0 + tz_1$ is computed on the fly. Each training step requires only: sampling t, z_0, z_1; computing z_t; evaluating $v_\theta(z_t,t)$; and computing the squared error against the constant target $z_1 - z_0$. No ODE solver is called at any point during training.

The linear interpolant path has an additional practical advantage. Because the target vector field is constant along each trajectory, the learned velocity field tends to produce nearly straight flow trajectories after training. Straight trajectories are easier to integrate accurately at inference time, requiring fewer ODE solver steps than CNFs trained by the original maximum likelihood approach where trajectories are curved (Liu et al., 2023). The training procedure described here applies to a single target distribution. The following section extends it to the credal mixture setting.

2.6.4 Conditioning on Auxiliary Information

The simulation-free training procedure of the previous section trains $v_\theta(z,t)$ to approximate a single target distribution. In the credal mixture framework we require K separate approximate posteriors $q_1^\theta, \ldots, q_K^\theta$, one for each prior π_k. A naive approach would train K independent flow matching models. As discussed in Section 2.5, this produces approximations of potentially uneven quality and does not satisfy the uniform approximation error condition required for Assumption 2.2.

The solution is to condition the velocity field on the prior index k directly. We extend:

$$v_\theta : \mathbb{R}^D \times [0,1] \times \{1,\ldots,K\} \to \mathbb{R}^D \tag{2.31}$$

written as $v_\theta(z,t,k)$. The conditioning is implemented by associating each prior index with a learned embedding vector $e_k \in \mathbb{R}^d$ and concatenating it with the time embedding before passing to the network:

$$\tilde{t}_k = [\psi(t) \,\|\, e_k] \in \mathbb{R}^{2d} \tag{2.32}$$

where $\psi : \mathbb{R} \to \mathbb{R}^d$ is a sinusoidal time embedding function. The embedding vectors $e_1, \ldots, e_K$ are learned jointly with the network parameters θ during training, adding only $K \times d$ parameters to the model, negligible compared to the total parameter count of the velocity field network.

The conditional flow matching objective extends naturally to the multi-prior setting. For each training step, a prior index $k \sim \mathcal{U}\{1,\ldots,K\}$ is sampled alongside the time and endpoint samples, and the target posterior $p_1 = p(\theta \mid x, \pi_k)$ is used to draw z_1:

$$\mathcal{L}_{\mathrm{CFM}}(\theta) = \mathbb{E}_{k,t,z_0,z_1 \sim p(\cdot|x,\pi_k)} \left\| v_\theta(z_t,t,k) - (z_1 - z_0) \right\|^2 \tag{2.33}$$

The velocity field backbone is shared across all K components; only the prior embedding e_k differentiates behaviour under each prior. All K approximate posteriors are therefore trained within a common optimisation framework with identical network capacity, directly satisfying the uniformity condition of Assumption 2.2.

This shared training framework has an important consequence for the credal mixture guarantee. Because all components share the same backbone, differences in approximation quality across priors arise only from genuine differences in posterior complexity rather than from differences in model capacity or training dynamics. The uniform approximation error bound $\varepsilon_k \leq \varepsilon$ of Assumption 2.2 is therefore a reasonable assumption under this architecture in a way that it would not be under independent training. The full pipeline is now in place: a single flow matching model with prior conditioning produces K approximate posteriors, and mixture weights are selected by minimising held-out NLL on $\mathcal{X}$. The theoretical properties of the resulting credal mixture are characterised by Theorem 2.12 in the following section.

2.7 The KL Convexity Guarantee Under Flow Approximation

The preceding sections have established the theoretical foundation for credal mixture inference and the computational machinery for realising it. We now ask whether the exact guarantee of Theorem 2.6 survives when posteriors are approximated by flow matching models and weights are selected from finite held-out data.

2.7.1 From Exact to Approximate Posteriors

Three things change in practice relative to Theorem 2.6. First, each posterior is replaced by a flow-based approximation q_k^θ. Second, mixture weights are selected by minimising held-out NLL on observations $x_1, \ldots, x_N \in \mathcal{X}$. Third, the theoretical objects of interest become predictive densities on $\mathcal{X}$ rather than posteriors on Θ.

Definition 2.10 (Predictive Densities)**.** The predictive density under q_k^* is $m_k(x) = \int \mathcal{L}(x \mid \theta)\, q_k^*(\theta)\, d\theta$; under q_k^θ it is $m_k^\theta(x) = \int \mathcal{L}(x \mid \theta)\, q_k^\theta(\theta)\, d\theta$; the mixture predictive is $m_w^\theta(x) = \sum_{k=1}^K w_k\, m_k^\theta(x)$.

The held-out NLL objective is $\hat{R}_N(w) = -\frac{1}{N} \sum_{i=1}^N \log m_w^\theta(x_i)$ with empirical minimiser $\hat{w}^N = \arg\min_w \hat{R}_N(w)$.

Two assumptions are required.

Assumption 2.1 (Uniform Empirical Risk Control)**.** The class $\{-\log m_w^\theta : w \in \Delta_{K-1}\}$ is Glivenko-Cantelli under P:

$$\sup_{w \in \Delta_{K-1}} \left| \hat{R}_N(w) - R(w) \right| \xrightarrow{\text{a.s.}} 0 \quad \text{as } N \to \infty \tag{2.34}$$

where $R(w) = \mathrm{KL}(P \| m_w^\theta) + c(P)$ and $c(P)$ does not depend on w.

Assumption 2.2 (Uniform Transfer Control)**.** There exists $\eta \geq 0$ such that for all k:

$$\mathrm{KL}(P \| m_k^\theta) \leq \mathrm{KL}(P \| m_k) + \eta \tag{2.35}$$

The shared-backbone architecture of Section 2.6 is designed to satisfy this by ensuring all K components are trained with comparable capacity.

Lemma 2.11 (Empirical Minimiser Excess Risk)**.** *Under Assumption 2.1, the empirical minimiser $\hat{w}^N$ satisfies:*

$$\mathrm{KL}(P \| m_{\hat{w}^N}^\theta) \leq \inf_{w \in \Delta_{K-1}} \mathrm{KL}(P \| m_w^\theta) + \delta_N \tag{2.36}$$

where $\delta_N = 2 \sup_{w \in \Delta_{K-1}} |\hat{R}_N(w) - R(w)| \xrightarrow{\text{a.s.}} 0$ *as* $N \to \infty$.

Proof. Since $\hat{w}^N$ minimises $\hat{R}_N$ over Δ_{K-1}, for every $w \in \Delta_{K-1}$ we have $\hat{R}_N(\hat{w}^N) \leq \hat{R}_N(w)$. Using the uniform convergence bound:

$$\begin{aligned} R(\hat{w}^N) &\leq \hat{R}_N(\hat{w}^N) + \sup_{u \in \Delta_{K-1}} |\hat{R}_N(u) - R(u)| \\ &\leq \hat{R}_N(w) + \sup_u |\hat{R}_N(u) - R(u)| \\ &\leq R(w) + 2 \sup_u |\hat{R}_N(u) - R(u)| \end{aligned} \tag{2.37}$$

Taking the infimum over w on the right and substituting $R(w) = \mathrm{KL}(P \| m_w^\theta) + c(P)$ with $c(P)$ constant in w yields the result. That $\delta_N \to 0$ almost surely follows directly from Assumption 2.1. $\square$

2.7.2 The Approximate Mixture Bound

Theorem 2.12 (Approximate Mixture Bound)**.** *Under Assumptions 2.1 and 2.2:*

$$\mathrm{KL}(P \| m_{\hat{w}^N}^\theta) \leq \min_k \mathrm{KL}(P \| m_k) + \eta + \delta_N \tag{2.38}$$

where $\delta_N \xrightarrow{\text{a.s.}} 0$ *as* $N \to \infty$.

Proof. The proof chains three inequalities. First, apply Lemma 2.11:

$$\mathrm{KL}(P \| m_{\hat{w}^N}^\theta) \leq \inf_{w \in \Delta_{K-1}} \mathrm{KL}(P \| m_w^\theta) + \delta_N \tag{2.39}$$

Second, apply Theorem 2.6 to the approximate family $\{m_k^\theta\}$:

$$\inf_{w\in\Delta_{K-1}} \mathrm{KL}(P\|m_w^\theta) \le \min_{k=1}^{K} \mathrm{KL}(P\|m_k^\theta) \tag{2.40}$$

Third, apply Assumption 2.2 to each component:

$$\min_{k=1}^{K} \mathrm{KL}(P\|m_k^\theta) \le \min_{k=1}^{K} [\mathrm{KL}(P\|m_k)+\eta] = \min_{k=1}^{K} \mathrm{KL}(P\|m_k)+\eta \tag{2.41}$$

Combining the three displays yields the result. □

Corollary 2.13 (Recovery of Exact Guarantee)**.** *As* $\eta \to 0$ *and* $N\to\infty$*:* $\mathrm{KL}(P\|m_{\hat{w}^N}^\theta)\to \min_k \mathrm{KL}(P\|m_k)$.

Proposition 2.14 (Sufficient Condition for Transfer Control)**.** *If* $\varepsilon_k \le \varepsilon$ *for all* k*,* $dP/dq_k^* \le \Gamma$ q_k^**-a.e., and* $|\log(q_k^*/q_k^\theta)| \le M$ *uniformly, then Assumption 2.2 holds with* $\eta = \Gamma C_M\sqrt{2\varepsilon}$ *where* $C_M = \sup_{r\in[e^{-M},e^M]} r|\log r|/|r-1|$.

Proof. Fix k and write $b=q_k^*$, $c=q_k^\theta$ for brevity. Observe that:

$$\mathrm{KL}(P\|c)-\mathrm{KL}(P\|b) = \int \log\frac{b(\theta)}{c(\theta)}\,dP(\theta) \tag{2.42}$$

Condition (2) gives $dP \le \Gamma\, dq_k^* = \Gamma\, b\, d\theta$, so:

$$|\mathrm{KL}(P\|c)-\mathrm{KL}(P\|b)| \le \Gamma\int \left|\log\frac{b(\theta)}{c(\theta)}\right| b(\theta)\,d\theta \tag{2.43}$$

Setting $r(\theta)=b(\theta)/c(\theta)$, condition (3) gives $r\in[e^{-M},e^M]$ pointwise, so $r|\log r| \le C_M|r-1|$. Therefore:

$$\int \left|\log\frac{b}{c}\right| b\,d\theta = \int r|\log r|\,c\,d\theta \le C_M\int |b-c|\,d\theta = 2C_M\|b-c\|_{\mathrm{TV}} \tag{2.44}$$

Pinsker's inequality and condition (1) give $\|b-c\|_{\mathrm{TV}} \le \sqrt{\varepsilon_k/2} \le \sqrt{\varepsilon/2}$. Combining:

$$\mathrm{KL}(P\|m_k^\theta) \le \mathrm{KL}(P\|m_k) + \Gamma C_M\sqrt{2\varepsilon} \tag{2.45}$$

for all k, so Assumption 2.2 holds with $\eta = \Gamma C_M\sqrt{2\varepsilon}$. □

Chapter 3

Methodology

3.1 Research Design and Hypotheses

This chapter describes the experimental framework used to investigate the central question of this thesis: whether the KL convexity guarantee established in Theorem 2.6 is empirically visible when posteriors are approximated by flow matching models and mixture weights are selected by held-out negative log likelihood. The theoretical guarantee is exact and holds under the conditions of Theorem 2.12, but whether those conditions are satisfied in practice, and whether the resulting improvement over the best single component is large enough to be practically meaningful, are empirical questions that the theory cannot answer.

3.1.1 Research Hypotheses

The experiments are organised around three hypotheses, each corresponding to a different aspect of the theoretical framework.

H1 (Guarantee Visibility). In the prior-sensitive regime, the credal mixture selected by held-out NLL will achieve lower test NLL than the best single-prior component selected by the same held-out criterion. This is the empirical counterpart of Theorem 2.12: if the approximation error η and the estimation error δ_N are both small, the mixture should dominate or match the best single component.

H2 (Prior Sensitivity Requirement). In the prior-insensitive regime, the mean test NLL of the credal mixture will be within 0.005 nats of the best single-prior component, and the difference will not be statistically significant at the 0.05 level under a paired Wilcoxon signed-rank test across seeds. This corresponds to the null regime identified in Corollary 2.13: when all posteriors have converged, the credal mixture reduces to the best single component and provides no additional benefit. Confirming this null result is as important as confirming H1, since it validates the theory rather than merely the method.

H3 (Algorithm Comparison). Among the weight optimisation algorithms, the EM algorithm will achieve the lowest worst-case regret across seeds, defined as:

$$\max_{s} \left(\mathrm{NLL}_{\mathrm{alg},s} - \mathrm{NLL}_{\mathrm{oracle},s}\right) \tag{3.1}$$

where the oracle is the best prior on the test set and s indexes random seeds. Worst-case regret is the primary criterion because the theoretical guarantee is a worst-case bound, and a weight selection algorithm that performs well on average but fails badly on some seeds does not reliably realise the guarantee. This is also the criterion most directly comparable to the theoretical bound of Theorem 2.12.

3.1.2 Experimental Design

The hypotheses are tested using a two-regime experimental design. A regime is considered prior-sensitive if the predictive log-likelihood gap between the best and worst prior exceeds 0.01 nats on the validation set, a threshold chosen to reflect a practically meaningful difference in predictive performance. Under this definition, the low-data German Credit setting ($n = 75$, 25 parameters) is prior-sensitive: the gap between the best prior (Gaussian or Laplace) and the worst (GaussianWide) reaches 0.13 to 0.27 nats across seeds, well above the threshold. The full-data setting ($n = 700$) is prior-insensitive by the same criterion, with a maximum gap of 0.002 nats across all priors and seeds.

H1 and H2 are tested against the same dataset under different data regimes, ensuring that any difference in results is attributable to prior sensitivity rather than to differences in the problem structure or model. H3 is tested by comparing eight weight optimisation algorithms on the same held-out data under identical conditions. The datasets, model architecture, training procedure, weight optimisation algorithms, and evaluation protocol are described in the following sections.

3.2 Datasets and Problem Formulations

The experiments span three categories of problem: a real-world Bayesian classification task that serves as the primary evaluation setting, a real-world regression task that provides additional evidence under a different likelihood structure, and synthetic two-dimensional datasets used to visualise posterior geometry and assess flow quality. Each category serves a distinct role in the experimental narrative.

3.2.1 German Credit (Bayesian Logistic Regression)

The primary dataset is the German Credit dataset (Hofmann, 1994), a binary classification problem with 1000 observations and 24 input features, available from the UCI Machine Learning Repository. The task is to predict credit risk (good or bad) from features including loan amount, duration, credit history, and employment status. We use the dataset under a Bayesian logistic regression model, which gives a 25-dimensional parameter space (24 weights plus a bias term). The likelihood is:

$$\mathcal{L}(x \mid \theta) = \prod_{i=1}^{n} \sigma(y_i \theta^\top \tilde{x}_i) \tag{3.2}$$

where σ is the sigmoid function, $y_i \in \{-1, +1\}$ are labels, and $\tilde{x}_i$ are the feature vectors. Features are standardised to zero mean and unit variance before training.

This dataset is used in two regimes that directly correspond to the two-regime experimental design of Section 3.1. In the low-data regime, a random subsample of $n = 75$ observations is drawn as the training set, leaving the remaining 925 observations for validation and test. With 25 parameters and only 75 observations, the data-to-parameter ratio of 3:1 ensures that the likelihood does not dominate the prior, producing genuine prior sensitivity. In the full-data regime, the full $n = 700$ observations are used for training, with the remaining 300 for validation and test. At this data-to-parameter ratio of 28:1, the likelihood dominates and all reasonable priors produce essentially equivalent posteriors. This regime serves as the null control for H2.

Experiments in the low-data regime are repeated across 5 random seeds, where each seed determines a different random subsample of the training data. This allows us to assess whether the identity of the best prior is stable across data subsamples, which is a prerequisite for the prior sensitivity claim.

3.2.2 Minnesota Radon

The Minnesota Radon dataset (Gelman and Hill, 2007) contains radon measurements from 919 houses across 85 counties in Minnesota. We use the two-dimensional formulation which models log radon concentration as a function of floor level (basement or first floor) using a Bayesian linear regression model with a two-dimensional parameter space (intercept and slope). The likelihood is:

$$\mathcal{L}(x \mid \theta) = \prod_{i=1}^{n} \mathcal{N}(y_i \mid \theta^\top \tilde{x}_i, \sigma^2) \tag{3.3}$$

with σ^2 fixed. The dataset contains approximately 620 training observations after the train-test split, giving a data-to-parameter ratio of 310:1. This high ratio means the dataset falls in the prior-insensitive regime by the criterion of Section 3.1, and is used as a supplementary null control alongside the full-data German Credit setting. Three priors are used: Gaussian, GaussianNarrow, and Student-T. Experiments are repeated across 10 random seeds.

3.2.3 Old Faithful

The Old Faithful dataset contains 272 observations of eruption duration and waiting time between eruptions for the Old Faithful geyser in Yellowstone National Park (Azzalini and Bowman, 1990). We use the two-dimensional formulation with eruption duration and waiting time as the two variables, modelled under a Bayesian Gaussian mixture framework. The dataset is well known for its bimodal structure, with two clearly separated clusters corresponding to short and long eruption cycles, making it a useful diagnostic for whether the flow can represent multimodal posteriors on real data.

With only 218 training observations after the train-test split and a two-dimensional parameter space, the data-to-parameter ratio is moderate. However, the strong bimodal structure of the data means the likelihood is highly informative about the cluster locations, and prior sensitivity is expected to be low. This dataset therefore serves as an additional supplementary experiment rather than a primary evaluation setting. Note that the critic-based weight optimisation used in initial experiments on this dataset produced NaN weights, confirming the instability of the adversarial approach identified in Section 3.5 and motivating the switch to direct NLL-based weight optimisation.

3.2.4 Synthetic Two-Dimensional Datasets

Three synthetic two-dimensional datasets are used to visualise posterior geometry, assess the quality of the flow-based approximations, and evaluate the credal mixture on problems with known ground truth density structure.

Eight-Ring. A mixture of eight Gaussian distributions arranged in a ring pattern in $\mathbb{R}^2$. This dataset tests whether the flow model can represent multimodal posteriors with strong rotational structure. The eight modes are evenly spaced on a circle of radius 3, each with standard deviation 0.3.

Two-Arm Spirals. Two interleaved spiral arms in $\mathbb{R}^2$, generating a highly non-convex density with long-range correlations between dimensions. This tests whether the flow can capture complex topological structure that mean field approximations cannot represent.

Two Moons. Two crescent-shaped clusters offset from each other in $\mathbb{R}^2$, a standard benchmark for approximate inference methods. The non-linear decision boundary and bimodal structure make it a useful diagnostic for posterior approximation quality.

For the synthetic datasets, the target distribution is the dataset density itself rather than a Bayesian posterior the flow is trained to approximate the data generating distribution under five different priors acting as additive perturbations. Five priors are used: Gaussian, GaussianNarrow, Laplace, Student-T, and RingMixture. Experiments are repeated across 10 random seeds. Evaluation uses kernel density estimation negative log likelihood (KDE-NLL) rather than predictive log likelihood, since there is no likelihood function in the Bayesian sense. These results are supplementary and serve primarily to visualise the flow approximations and the credal mixture geometry.

3.2.5 Credal Sets

The credal set for each problem is constructed by specifying K prior distributions over the model parameters θ. Table 3.1 summarises the priors used for each dataset.

The priors in each credal set are chosen to represent the range of beliefs a practitioner might reasonably hold before observing data. They vary along two axes: tail behaviour (from light-tailed Gaussian to heavy-tailed Cauchy) and scale (from the informative GaussianNarrow to the weakly informative GaussianWide). No single prior

Table 3.1: Credal sets used for each experimental setting. All Gaussian priors are isotropic. σ denotes the standard deviation, ν the degrees of freedom for Student-T, and b the scale for Laplace.

Dataset	Prior	Parameters	Notes
German Credit	Gaussian	$\sigma = 1.0$	Standard
	Laplace	$b = 1.0$	Heavy-tailed
	Cauchy	$\gamma = 1.0$	Very heavy
	Student-T	$\nu = 3$	Moderate tails
	GaussianWide	$\sigma = 5.0$	Weakly informative
Radon MN	Gaussian	$\sigma = 1.0$	Standard
	GaussianNarrow	$\sigma = 0.3$	Informative
	Student-T	$\nu = 3$	Moderate tails
Old Faithful	Gaussian	$\sigma = 1.0$	Standard
	GaussianNarrow	$\sigma = 0.3$	Informative
	Student-T	$\nu = 3$	Moderate tails
Synthetic	Gaussian	$\sigma = 1.0$	Standard
	GaussianNarrow	$\sigma = 0.5$	Informative
	Laplace	$b = 1.0$	Heavy-tailed
	Student-T	$\nu = 3$	Moderate tails
	RingMixture	–	Multimodal

is known to be correct a priori, which is precisely the motivation for the credal mixture approach.

3.3 Model Architecture

All experiments use a single shared conditional velocity field, implemented as a neural network called `VelocityResNet`. The key design principle is that one model serves all K priors simultaneously: rather than training K separate networks, a single network learns to approximate the posterior under each prior by conditioning on a discrete prior index. This shared backbone is what makes the credal mixture computationally tractable and ensures the uniform approximation quality required by Assumption 2.2.

At a high level, the network takes three inputs at each training step: the current position $z \in \mathbb{R}^D$ in parameter space, the current time $t \in [0,1]$ along the flow trajectory, and the prior index $k \in \{1,\ldots,K\}$ identifying which prior is active. It outputs a velocity vector $v_\theta(z,t,k) \in \mathbb{R}^D$ that tells a particle at position z at time t which direction to move in order to travel from the base distribution toward the posterior under prior π_k. The network architecture is summarised in Table 3.2.

Table 3.2: Common architecture and optimiser settings across all experiments.

Parameter	Value	Role
Architecture		
Model	VelocityResNet	Velocity field backbone
Hidden dimension	256	Width of each layer
Depth	6 FiLM blocks	Number of processing stages
Time embedding dim	64	Encoding of t
Prior embedding dim	16	Encoding of prior index k
Dropout	0.05	Regularisation
Activation	SiLU	Non-linearity
Flow Matching		
Path	Linear interpolant	Straight-line trajectories
Base distribution	Prior π_k (Bayesian); $\mathcal{N}(0,I)$ (synthetic)	Starting distribution
Time sampling	$\mathcal{U}[0,1]$	Uniform over trajectory
ODE solver (inference)	Dormand-Prince	Adaptive step integration
Optimiser		
Optimiser	Adam	Gradient-based training
Learning rate	1×10^{-3}	Step size
Weight decay	1×10^{-5}	L2 regularisation
Batch size	512	Samples per update

Figure 3.1 provides an overview of the full training pipeline before each component is described in detail.

3.3.1 Input Processing

The network receives three inputs that must be encoded into a form suitable for a neural network before any processing can begin.

The position $z \in \mathbb{R}^D$ represents the current location of a particle in the parameter space Θ. At $t = 0$ it is drawn from the source distribution, which is the prior π_k for Bayesian experiments and $\mathcal{N}(0,I)$ for synthetic experiments, and by $t = 1$ it should have been transported to the approximate posterior. The position is projected into the hidden dimension by a two-layer MLP with SiLU activations:

$$h_0 = W_2\,\sigma(W_1 z + b_1) + b_2 \in \mathbb{R}^{256} \tag{3.4}$$

where σ denotes the SiLU activation function $\sigma(x) = x \cdot \text{sigmoid}(x)$, which is smooth

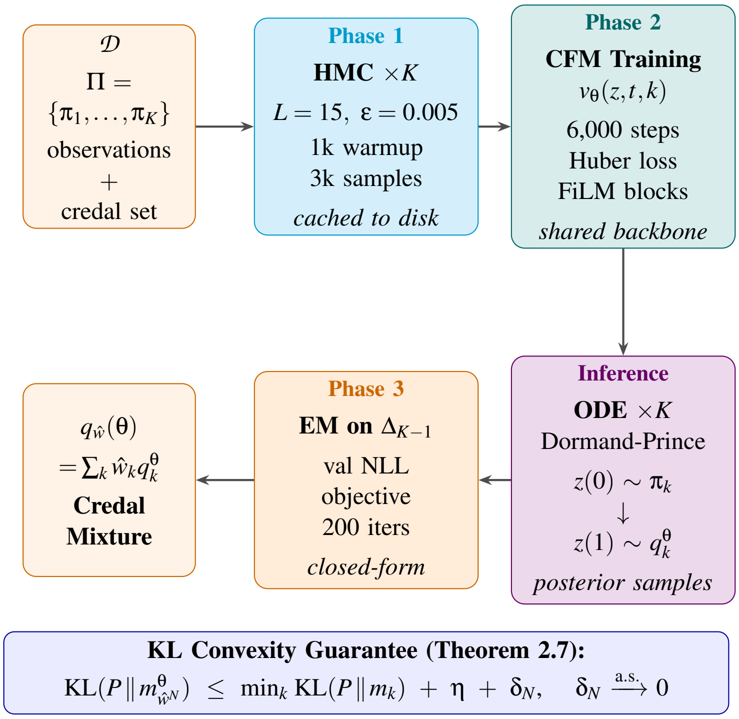

Figure 3.1: Full pipeline for robust credal variational inference. Phase 1 generates and caches HMC posterior samples per prior. Phase 2 trains a shared conditional flow matching model. At inference, K ODE integrations produce approximate posteriors. Phase 3 selects optimal mixture weights via EM on the simplex.

and avoids the dying neuron problem of ReLU. This initial projection h_0 is the starting hidden state that is then refined by the sequence of FiLM blocks.

The time $t \in [0, 1]$ and the prior index k are encoded separately and then combined into a single conditioning vector, described in the following subsections.

3.3.2 Time Embedding

The time $t \in [0, 1]$ needs to be encoded in a way that allows the network to distinguish fine-grained differences in position along the trajectory, particularly near $t = 0$ and $t = 1$ where the distribution changes most rapidly. A single scalar t is too limited for this: a neural network struggles to represent high-frequency variation from a raw scalar input.

The solution is a sinusoidal embedding that maps t to a vector of sine and cosine values at different frequencies, similar to the positional encoding used in transformer models (Vaswani et al., 2017). For $n_f = 12$ frequency bands $\mathbf{f} = [2^0, 2^1, \ldots, 2^{n_f-1}]$:

$$\psi(t) = \mathrm{MLP}\Big(\big[t,\ \sin(2\pi t\cdot\mathbf{f}),\ \cos(2\pi t\cdot\mathbf{f})\big]\Big) \in \mathbb{R}^{64} \tag{3.5}$$

where $[\cdot]$ denotes concatenation and the MLP has two hidden layers with SiLU activations. The sine and cosine terms at different frequencies act as a bank of clocks running at different speeds: low frequencies capture the coarse position along the trajectory while high frequencies capture fine detail. The MLP then compresses this $1 + 2n_f = 25$ dimensional input into a 64-dimensional time embedding.

3.3.3 Prior Embedding and Conditioning Vector

The prior index $k \in \{1, \ldots, K\}$ is a discrete label identifying which prior is active. Neural networks cannot process discrete labels directly, so the index is mapped to a continuous vector via a learned embedding table. This is implemented as `nn.Embedding`, which maintains a lookup table of K vectors $\{e_1, \ldots, e_K\} \subset \mathbb{R}^{16}$, one per prior. The embedding for prior k is simply e_k, which is learned jointly with all other network parameters during training.

The intuition is that the embedding learns to place priors that produce similar posteriors close together in the 16-dimensional embedding space, and priors that produce very different posteriors far apart. This gives the network a geometric representation of the relationships between priors that it can use to modulate its behaviour appropriately.

The time embedding and prior embedding are concatenated to form the conditioning vector:

$$c_k(t) = [\psi(t) \parallel e_k] \in \mathbb{R}^{80} \tag{3.6}$$

This single 80-dimensional vector encodes both when the particle is in its trajectory and which prior it is flowing toward. It is passed into every FiLM block to modulate the network's behaviour at each processing stage.

3.3.4 FiLM-Conditioned Residual Blocks

The core of the network is a stack of $L = 6$ Feature-wise Linear Modulation (FiLM) blocks (Perez et al., 2018). Each block takes the current hidden state $h \in \mathbb{R}^{256}$ and the conditioning vector $c_k(t)$ as inputs, and produces an updated hidden state $h' \in \mathbb{R}^{256}$.

FiLM conditioning works by learning to scale and shift the hidden state differently for each prior and time. Two small linear layers produce a scale vector γ and a shift vector β from the conditioning vector:

$$\gamma = W_\gamma c_k(t) + b_\gamma \in \mathbb{R}^{1024}, \qquad \beta = W_\beta c_k(t) + b_\beta \in \mathbb{R}^{1024} \tag{3.7}$$

These are applied elementwise to the output of an inner MLP acting on the normalised hidden state:

$$h' = \gamma \odot \sigma(W_2\, \sigma(W_1\, \mathrm{LN}(h) + b_1) + b_2) + \beta + h \tag{3.8}$$

where LN denotes Layer Normalisation (Ba et al., 2016), $\odot$ denotes elementwise multiplication, and the final $+h$ is the residual skip connection. The inner MLP has hidden dimension $4 \times 256 = 1024$, which gives the network enough capacity to compute complex transformations of the hidden state before the FiLM modulation is applied.

The residual connection $+h$ is important for two reasons. First, it allows gradients to flow directly from the output to the input during backpropagation, which helps training deep networks. Second, it means the block only needs to learn the difference between the input and the desired output rather than the output from scratch, which is a much easier learning problem. The FiLM parameters γ and β then determine how much and in which direction this residual is adjusted for each prior and time. Table 3.3 summarises the FiLM block configuration.

Table 3.3: FiLM block configuration.

Parameter	Value	Role
Input dimension	256	Hidden state width
Inner MLP hidden dim	1024	$4\times$ hidden expansion
Conditioning dimension	80	$[\psi(t) \| e_k]$
Normalisation	LayerNorm	Stabilises training
Activation	SiLU	Smooth non-linearity
Dropout	0.05	Prevents overfitting
Residual connection	Yes	Gradient flow
Number of blocks	6	Processing depth

3.3.5 Output Head and Velocity Prediction

After passing through all $L = 6$ FiLM blocks, the final hidden state $h_L \in \mathbb{R}^{256}$ is passed through a single linear layer to produce the velocity prediction:

$$v_\theta(z,t,k) = W_{\text{head}}\, h_L + b_{\text{head}} \in \mathbb{R}^D \tag{3.9}$$

This velocity vector tells a particle at position z at time t under prior k exactly how to move. When the network is well trained, integrating this velocity field from $t = 0$ to $t = 1$ using an ODE solver transports a sample from the prior π_k to a sample from the approximate posterior q_k^θ under prior π_k, with $\mathcal{N}(0, I)$ used as the source for the synthetic 2D experiments where no likelihood is defined.

3.3.6 Dataset-Specific Configurations

The input dimension D and number of priors K vary by dataset. All other parameters are fixed as in Table 3.2. Table 3.4 summarises the dataset-specific configurations.

Table 3.4: Dataset-specific configurations. All other parameters follow Table 3.2.

Dataset	Input dim D	Priors K	Training steps
German Credit (low)	25	5	20,000
German Credit (full)	25	5	20,000
Radon MN	2	3	10,000
Old Faithful	2	3	10,000
Eight-Ring	2	5	10,000
Two-Arm Spirals	2	5	10,000
Two Moons	2	5	10,000

The parameter count of the network scales with D only through the input projection and output head layers, which are small relative to the total. For German Credit with $D = 25$ the total parameter count is approximately 1.2 million, and for the 2D datasets it is approximately 1.1 million. The difference is negligible, confirming that the shared architecture provides comparable capacity across all experimental settings.

3.4 Training Procedure

The two-phase training procedure is identical across all experiments, with the following dataset-specific variations. For German Credit, HMC is run separately for the low-data ($n = 75$) and full-data ($n = 700$) training sets, producing different posterior caches for each regime. For Minnesota Radon and Old Faithful, HMC is run on the respective training splits with the same hyperparameters. For the synthetic 2D datasets, posterior samples are drawn directly from the dataset distribution under each prior perturbation rather than from an HMC chain, since there is no likelihood function in the Bayesian sense. The number of training steps varies by input dimensionality as shown in Table 3.4: 20,000 steps for the 25-dimensional German Credit problems and 10,000 steps for all 2-dimensional problems.

3.4.1 Phase 1: Posterior Sample Generation via HMC

For each prior π_k in the credal set, posterior samples are generated using a self-contained Hamiltonian Monte Carlo sampler implemented in PyTorch, which uses automatic differentiation to compute the gradient of the log posterior without requiring manual derivation. HMC is chosen because it uses gradient information to make large, informed proposals in parameter space, producing lower autocorrelation and faster mixing than random walk methods such as Metropolis-Hastings (Neal, 2011). The full procedure is illustrated in Figure 3.2.

Each HMC transition proceeds as follows. A momentum vector $p \sim \mathcal{N}(0, I)$ is sampled, and the Hamiltonian is defined as:

$$H(\theta, p) = -\log \pi_k(\theta) - \log \mathcal{L}(x \mid \theta) + \frac{1}{2}\|p\|^2 \tag{3.10}$$

The leapfrog integrator then evolves (θ, p) for L steps of size ε, and the proposal is accepted or rejected by a Metropolis correction based on the change in Hamiltonian. The HMC configuration is summarised in Table 3.5.

Table 3.5: HMC configuration for posterior sample generation. The same settings are used for all priors and datasets.

Parameter	Value	Role
Samples per prior	3,000	Size of posterior cache
Warmup steps	1,000	Burn-in period
Step size ε	0.005	Leapfrog step size
Leapfrog steps L	15	Steps per transition
Initialisation	$\theta = 0$	Starting point
Backend	PyTorch autograd	Gradient computation

The warmup period of 1,000 steps allows the chain to move away from the initialisation at zero and reach the high-probability region of the posterior before samples are retained. The 3,000 post-warmup samples are saved to disk as a cache and reused across all training runs, so HMC is run only once per prior per dataset configuration. During training, the flow model draws batches from this cache via random sampling with replacement.

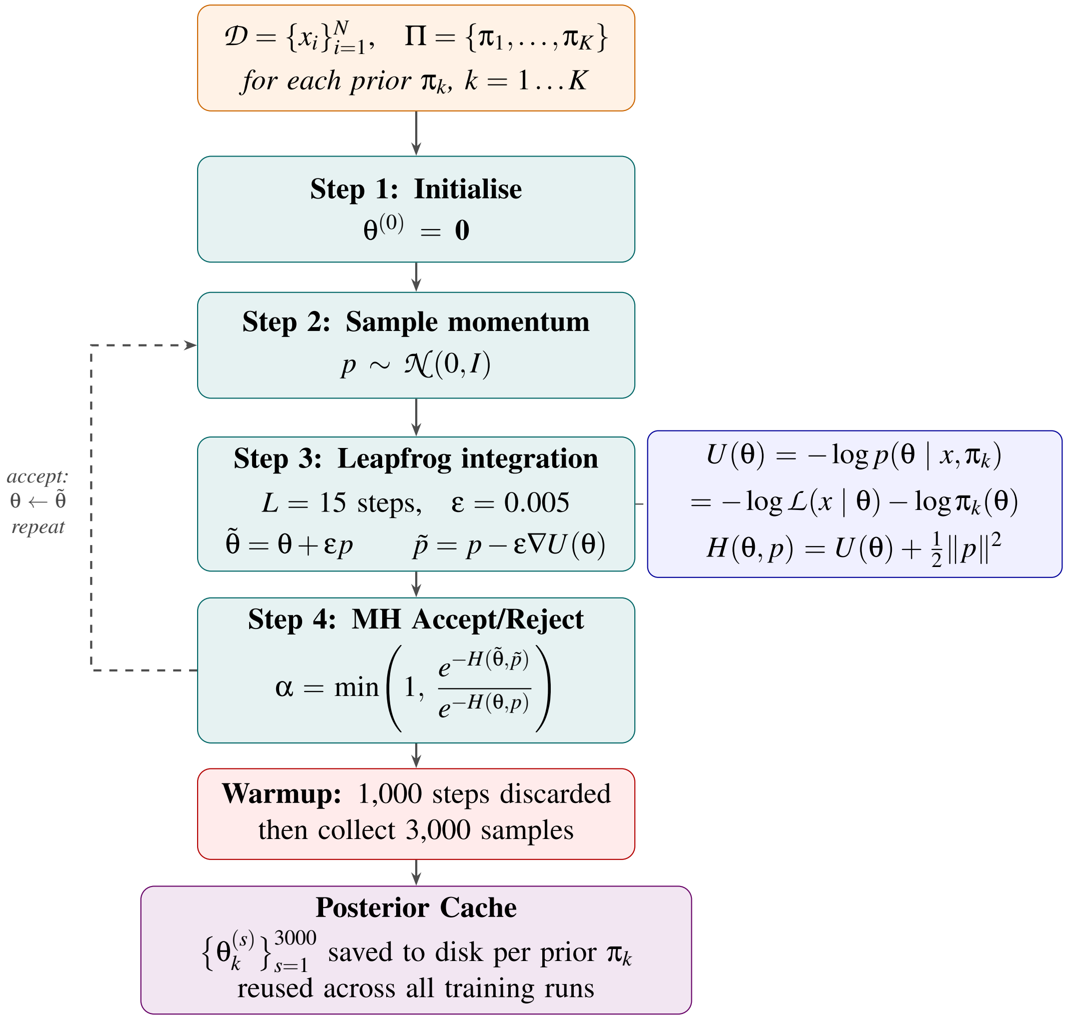

Figure 3.2: Phase 1: HMC posterior sampling. For each prior $\pi_k \in \Pi$, 3,000 posterior samples are generated after a 1,000-step warmup and cached to disk for reuse across all training runs.

3.4.2 Phase 2: Conditional Flow Matching Training

With posterior samples cached for each prior, the velocity field v_θ is trained using the conditional flow matching objective derived in Section 2.6. The full training loop is illustrated in Figure 3.3, and each step is described in detail below.

At each training step the following procedure is followed:

1. Sample a prior index $k \sim \mathcal{U}\{1, \dots, K\}$ uniformly at random
2. Draw a target sample $z_1 \sim q_k^*$ from the HMC posterior cache for prior π_k
3. Draw a source sample $z_0 \sim \pi_k$ from the prior distribution directly, and clip to $[-C, C]$ with $C = 15.0$ to prevent extreme outliers from heavy-tailed priors such

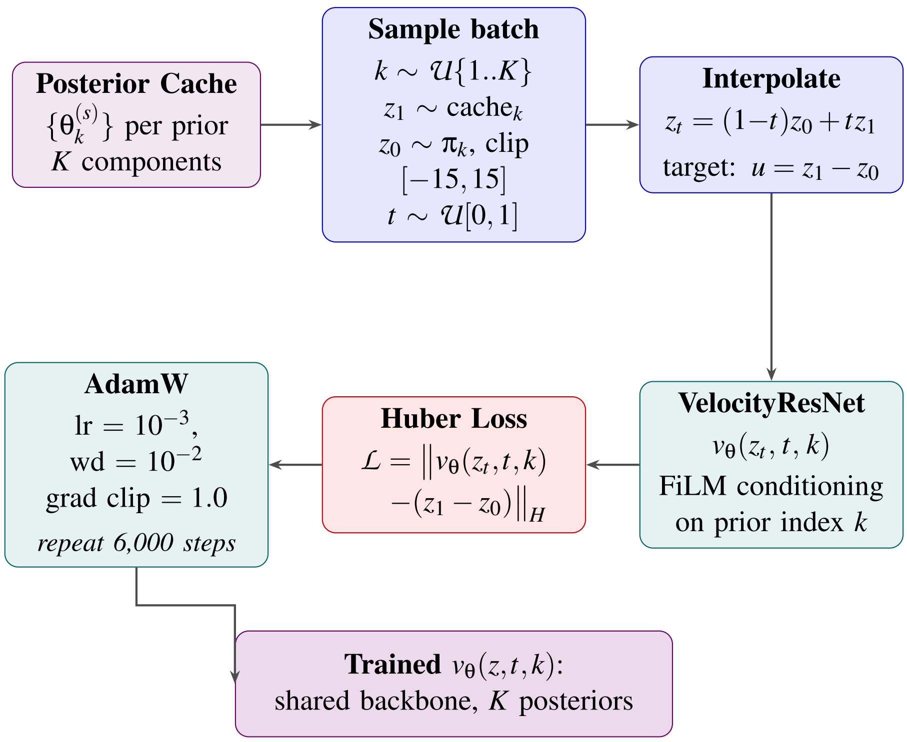

Figure 3.3: Phase 2: Conditional Flow Matching training. A single shared `VelocityResNet` is trained jointly across all K priors using the simulation-free CFM objective.

as Cauchy destabilising training

4. Sample a time $t \sim \mathcal{U}[\varepsilon, 1-\varepsilon]$ with $\varepsilon = 10^{-4}$ to avoid boundary effects at $t = 0$ and $t = 1$
5. Compute the interpolant $z_t = (1-t)z_0 + tz_1$ and the target velocity $u = z_1 - z_0$
6. Compute the Huber loss between the predicted and target velocity:

$$\ell(\theta) = \text{HuberLoss}(v_\theta(z_t, t, k),\ z_1 - z_0) \tag{3.11}$$

7. Backpropagate and update θ with AdamW, clipping gradients to maximum norm 1.0

The source distribution is the prior π_k rather than a standard Gaussian $\mathcal{N}(0, I)$. This is the natural choice for Bayesian inference: the flow only needs to learn the update from the prior to the posterior, which is a much smaller distributional shift than transporting from a standard Gaussian. For well-specified priors such as Gaussian with $\sigma = 1.0$, the prior and posterior are already close, and the flow learns a small correction. For misspecified priors such as GaussianWide with $\sigma = 1.5$, the shift is larger and more training steps are required.

The Huber loss is used in place of the mean squared error of equation (2.30) for robustness to outlier velocities. For heavy-tailed priors such as Cauchy and Student-T, occasional extreme source samples z_0 can produce very large target velocities $z_1 - z_0$ even after clipping. The Huber loss treats errors below a threshold δ as squared loss and errors above it as linear loss, reducing the influence of these outlier velocities on the gradient without discarding them entirely. The threshold is set to $\delta = 5.0$. This substitution is a practical heuristic: it breaks the exact gradient equivalence of Theorem 2.9, which is proved for the squared error objective, but the equivalence is preserved approximately for velocities below the threshold where the two losses agree.

3.4.3 Training Details and Convergence

Training is run for a fixed number of steps rather than to a convergence criterion, with 20,000 steps for the 25-dimensional German Credit problems and 10,000 steps for the 2-dimensional problems. These step counts were determined by monitoring the training loss and confirming that it had stabilised before the end of training. The training configuration is summarised in Table 3.6.

Table 3.6: Training configuration for the conditional flow matching model.

Parameter	Value	Role
Optimiser	AdamW	Weight update
Learning rate	1×10^{-3}	Step size
Weight decay	1×10^{-5}	L2 regularisation
Batch size	512	Samples per step
Gradient clip	1.0 (max norm)	Training stability
Loss function	Huber ($\delta = 1.0$)	Robust regression
Source clip	$C = 5.0$	Heavy-tail robustness
Time epsilon	$\varepsilon = 10^{-4}$	Boundary avoidance
Steps (2D)	10,000	
Steps (25D)	20,000	

All models are trained on a single device, with MPS used on Apple Silicon hardware and CUDA used where available, falling back to CPU otherwise. Each training run is seeded for reproducibility using the same seed for PyTorch, NumPy, and Python's random module. Different seeds produce different random subsamples of the training data (for the low-data German Credit regime) and different random draws from the posterior cache during training.

3.5 Credal Weight Optimisation

After training the conditional flow matching model, the K approximate posteriors $q_1^\theta, \ldots, q_K^\theta$ are combined into a credal mixture by optimising the mixture weights $w \in \Delta_{K-1}$ on a held-out validation set. This section describes the shared objective, the eight optimisation algorithms compared, and the practical implementation.

3.5.1 The Shared Objective

All eight algorithms optimise the same objective: minimise the held-out mixture negative log likelihood over the simplex:

$$\hat{w} = \underset{w \in \Delta_{K-1}}{\arg\min} \hat{R}_N(w) = -\frac{1}{N} \sum_{i=1}^{N} \log \sum_{k=1}^{K} w_k m_k^\theta(x_i) \tag{3.12}$$

where $x_1, \ldots, x_N$ are held-out validation observations and $m_k^\theta(x_i)$ is the predictive log-density of component k at observation x_i.

To avoid recomputing the predictive densities at every iteration, the matrix $\mathbf{L} \in \mathbb{R}^{N \times K}$ of log predictive densities is precomputed once before optimisation:

$$\mathbf{L}_{ik} = \log m_k^\theta(x_i), \quad i = 1, \ldots, N, \quad k = 1, \ldots, K \tag{3.13}$$

All subsequent computations use $\mathbf{L}$ directly, making each iteration cheap regardless of the complexity of the flow model. The mixture NLL for a given weight vector w is then computed using the log-sum-exp trick for numerical stability:

$$\hat{R}_N(w) = -\frac{1}{N} \sum_{i=1}^{N} \operatorname{logsumexp}_k \left(\mathbf{L}_{ik} + \log w_k \right) \tag{3.14}$$

The gradient of $\hat{R}_N(w)$ with respect to w_k is:

$$\frac{\partial \hat{R}_N}{\partial w_k} = -\frac{1}{N} \sum_{i=1}^{N} \frac{m_k^\theta(x_i)}{\sum_j w_j m_j^\theta(x_i)} = -\frac{1}{N} \sum_{i=1}^{N} r_{ik}(w) \tag{3.15}$$

where $r_{ik}(w)$ is the responsibility of component k for observation x_i under weights w:

$$r_{ik}(w) = \frac{w_k m_k^\theta(x_i)}{\sum_j w_j m_j^\theta(x_i)} \tag{3.16}$$

The responsibilities are computed in log space for numerical stability and are used by several of the algorithms below.

3.5.2 Optimisation Algorithms

Eight algorithms are compared, ranging from the naive baselines of uniform weighting and best-single selection to principled optimisation methods on the simplex. The algorithms are grouped by type.

Baseline methods. *Uniform* assigns equal weight $w_k = 1/K$ to all components without any optimisation. This is the simplest possible credal mixture and serves as a lower bound on performance.

Best-single selects the single component with the lowest held-out NLL and assigns it weight 1.0. Formally:

$$\hat{w} = e_{k^*}, \quad k^* = \arg\min_k \hat{R}_N(e_k) \tag{3.17}$$

where e_k is the k-th standard basis vector. This is the algorithm a practitioner would use if they selected a single prior by held-out validation, and it is the primary baseline against which the mixture algorithms are compared.

EM algorithm. Among the weight optimisation algorithms evaluated in this thesis, EM is the primary recommended method. It treats the mixture component index as a latent variable and alternates between computing responsibilities (E-step) and updating weights (M-step), with a closed-form update that requires no learning rate and is guaranteed to decrease the mixture NLL at every iteration (Dempster et al., 1977). The procedure is illustrated in Figure 3.4.

The E-step computes the responsibility of each component k for each observation x_i:

$$r_{ik}^{(t)} = \frac{w_k^{(t)} m_k^{\theta}(x_i)}{\sum_j w_j^{(t)} m_j^{\theta}(x_i)} \tag{3.18}$$

The M-step updates the weights as the mean responsibility across all observations:

$$w_k^{(t+1)} = \frac{1}{N} \sum_{i=1}^{N} r_{ik}^{(t)} \tag{3.19}$$

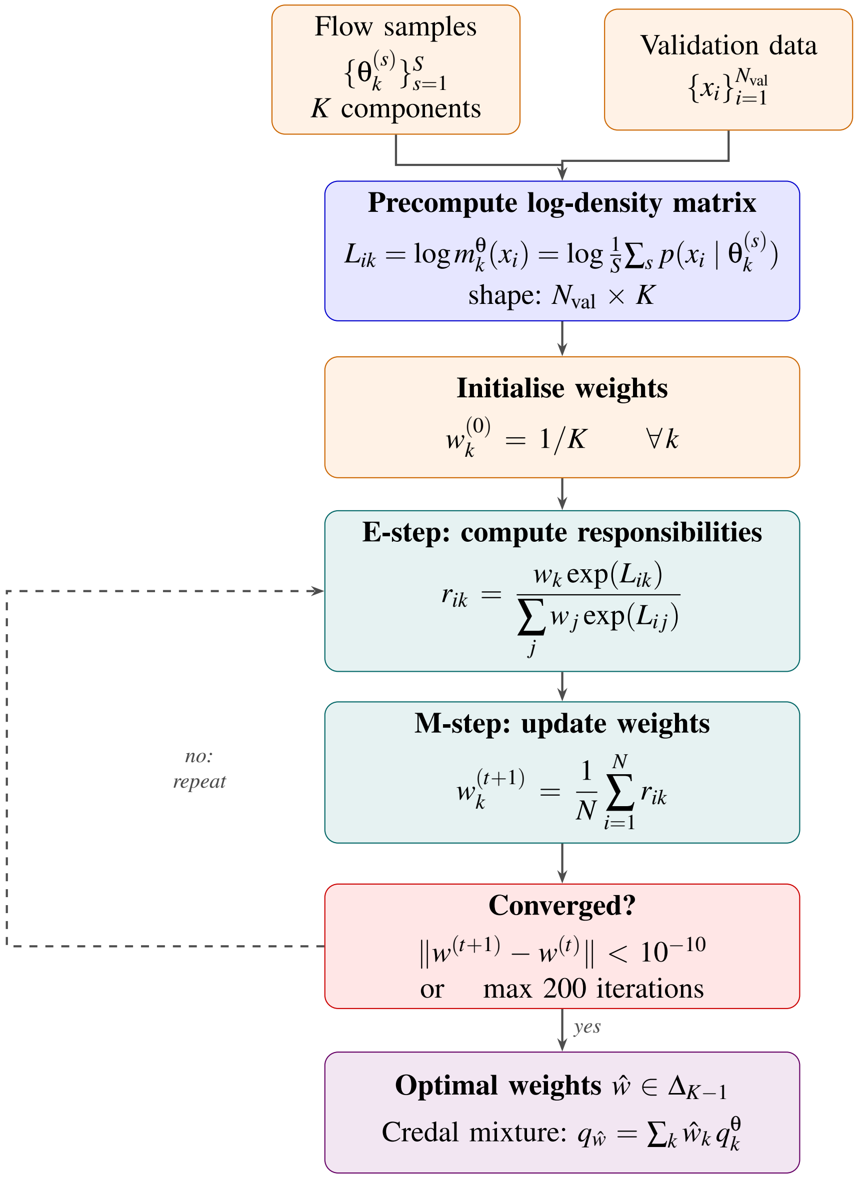

Figure 3.4: Phase 3: EM weight optimisation on Δ_{K-1}. The E-step and M-step alternate until convergence, producing optimal mixture weights $\hat{w}$ that minimise the held-out negative log-likelihood.

This update has a closed form, requires no learning rate, and is guaranteed to decrease the mixture NLL at every iteration. The algorithm runs for up to 200 iterations with early stopping when the change in NLL falls below 10^{-10}, and is initialised at uniform weights $w_k = 1/K$. EM consistently achieves the lowest worst-case regret among all algorithms evaluated in Chapter 4, making it the recommended default for credal weight selection in practice.

Gradient-based methods. *Projected gradient descent* (PGD) applies gradient descent with momentum followed by projection onto the simplex (Duchi et al., 2008). Starting from uniform weights:

$$v^{(t+1)} = \mu v^{(t)} + \nabla_w \hat{R}_N(w^{(t)}) \tag{3.20}$$

$$w^{(t+1)} = \Pi_{\Delta_{K-1}} \left(w^{(t)} - \alpha v^{(t+1)} \right) \tag{3.21}$$

where $\Pi_{\Delta_{K-1}}$ denotes projection onto the simplex, $\mu = 0.9$ is the momentum coefficient, and $\alpha = 0.05$ is the learning rate. The simplex projection is computed using the algorithm of Duchi et al. (2008) in $O(K \log K)$ time. The algorithm runs for 400 iterations.

Mirror descent with exponentiated gradients uses a multiplicative update that maintains positivity and the simplex constraint implicitly, without requiring projection:

$$w_k^{(t+1)} = \frac{w_k^{(t)} \exp\left(-\alpha \frac{\partial \hat{R}_N}{\partial w_k}(w^{(t)}) \right)}{\sum_j w_j^{(t)} \exp\left(-\alpha \frac{\partial \hat{R}_N}{\partial w_j}(w^{(t)}) \right)} \tag{3.22}$$

with learning rate $\alpha = 0.05$ and 400 iterations. Mirror descent is the natural algorithm for the simplex under the negative entropy geometry (Beck and Teboulle, 2003) but is known to be sensitive to the learning rate choice.

Frank-Wolfe. The Frank-Wolfe algorithm Frank and Wolfe (1956) avoids projection by using a linear oracle to find the best single component at each step and then mixing the current iterate with that component:

$$s^{(t)} = e_{k^*}, \quad k^* = \arg\min_k \nabla_w \hat{R}_N(w^{(t)})^\top e_k \tag{3.23}$$

$$w^{(t+1)} = (1 - \gamma^{(t)}) w^{(t)} + \gamma^{(t)} s^{(t)} \tag{3.24}$$

where the step size $\gamma^{(t)}$ is selected by a line search over $[0,1]$. The Frank-Wolfe algorithm has a natural sparse solution structure: it tends to concentrate weight on a small number of components, which mirrors the expected behaviour when one prior dominates. It runs for 200 iterations.

Coordinate pair search. The coordinate pair search algorithm randomly selects two components (a,b) at each iteration and finds the optimal redistribution of their combined mass $m = w_a + w_b$ by a 1D grid search:

$$(w_a^*, w_b^*) = \underset{t \in \{0, \frac{1}{G}, \dots, 1\}}{\arg\min} \; \hat{R}_N(w \text{ with } w_a = tm,\, w_b = (1-t)m) \tag{3.25}$$

with grid size $G = 50$, run for 300 iterations with random pair selection. This is a coordinate descent method that avoids computing the full gradient by solving a sequence of one-dimensional subproblems.

Grid search. Grid search exhaustively evaluates the NLL at all weight vectors on a regular grid over Δ_{K-1} with step size 0.05, returning the minimiser. For $K = 5$ this grid has $\binom{24}{4} = 10{,}626$ points. Grid search is guaranteed to find the global minimum within the grid resolution and serves as a near-exhaustive baseline. It is computationally feasible only for small K.

3.5.3 Algorithm Summary

Table 3.7 summarises all eight algorithms with their key hyperparameters.

Table 3.7: Credal weight optimisation algorithms. All algorithms optimise the same held-out NLL objective over Δ_{K-1}.

Algorithm	Type	Iterations	Hyperparameters	Constraint
Uniform	Baseline	–	–	Implicit
Best-single	Baseline	–	–	Vertex
EM	Closed-form	200	tol $= 10^{-10}$	Implicit
Proj. GD	Gradient	400	lr $= 0.05$, $\mu = 0.9$	Projection
Mirror descent	Gradient	400	lr $= 0.05$	Implicit
Frank-Wolfe	Conditional grad	200	line search	Implicit
Coord. pair	Coordinate	300	grid $= 50$	Implicit
Grid search	Exhaustive	–	step $= 0.05$	Exhaustive

All algorithms are initialised at uniform weights $w_k = 1/K$ and operate on the precomputed log-density matrix $\mathbf{L}$. The same validation set and the same $\mathbf{L}$ matrix are used for all algorithms, ensuring a fair comparison. Results are reported across multiple random seeds, where each seed determines a different training data subsample and a different draw from the HMC posterior cache.

Chapter 4

Results and Discussion

This chapter presents and interprets the empirical results across all experimental settings. Section 4.1 reports the primary result on the German Credit low-data regime, directly testing H1 and H3. Section 4.2 presents the null control on the full-data regime, testing H2. Section 4.3 presents the Minnesota Radon supplementary null control. Section 4.4 presents the architectural ablation validating the shared backbone. Section 4.5 presents the diagnostic analyses: prior sensitivity, weight stability, leave-one-prior-out, calibration, and optimiser convergence. Section 4.6 reports the synthetic 2D results. Section 4.7 summarises the hypothesis outcomes. Section 4.5.6 presents a direct speed comparison between flow matching and CNF training, validating the simulation-free computational advantage.

4.1 Primary Result: German Credit Low-Data Regime

The primary experimental setting is German Credit with $n = 75$ training observations and 25 parameters. Under this 3:1 data-to-parameter ratio the likelihood is insufficiently informative to overwhelm any reasonable prior, placing the experiment firmly in the prior-sensitive regime as defined in Section 3.1: the gap between the best and worst prior on the validation set reaches 0.13 to 0.27 nats across seeds, well above the 0.01 nat threshold.

4.1.1 Mean NLL and Regret

Table 4.1 reports the test NLL per seed and the mean across 5 seeds for all eight weight optimisation algorithms, together with the oracle best-prior NLL computed post-hoc on the test set. Table 4.2 reports the regret relative to the oracle at each seed, with the maximum and mean regret across seeds.

Table 4.1: Test NLL per seed and mean for all algorithms on German Credit low-data regime ($n = 75$, 5 seeds). Lower is better.

Algorithm	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Mean
EM	0.6307	0.6214	0.6169	0.5976	0.6461	**0.6225**
Proj. GD	0.6310	0.6215	0.6176	0.5972	0.6464	0.6227
Coord. pair	0.6312	0.6215	0.6175	0.5971	0.6468	0.6228
Grid search	0.6312	0.6215	0.6175	0.5971	0.6467	0.6228
Frank-Wolfe	0.6312	0.6215	0.6175	0.5971	0.6468	0.6228
Best-single	0.6333	0.6222	0.6175	0.5971	0.6500	0.6240
Mirror desc.	0.6380	0.6324	0.6258	0.6115	0.6546	0.6325
Uniform	0.6753	0.6735	0.6671	0.6473	0.6881	0.6703
Oracle	0.6307	0.6214	0.6169	0.5971	0.6461	0.6224

Table 4.2: Regret (NLL minus oracle NLL) per seed for all algorithms. Max and mean regret across 5 seeds reported. Lower is better.

Algorithm	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Max	Mean
EM	0.0000	0.0000	0.0000	0.0005	0.0000	**0.0005**	0.0001
Grid	0.0005	0.0001	0.0006	0.0000	0.0006	0.0006	0.0004
Proj. GD	0.0003	0.0001	0.0007	0.0001	0.0003	0.0007	0.0003
Coord. pair	0.0005	0.0001	0.0006	0.0000	0.0007	0.0007	0.0004
Frank-Wolfe	0.0005	0.0001	0.0006	0.0000	0.0007	0.0007	0.0004
Best-single	0.0026	0.0008	0.0006	0.0000	0.0039	0.0039	0.0016
Mirror desc.	0.0073	0.0110	0.0089	0.0144	0.0085	0.0144	0.0100
Uniform	0.0446	0.0521	0.0502	0.0502	0.0420	0.0521	0.0478

4.1.2 H1: Guarantee Visibility

All principled mixture algorithms (EM, Proj. GD, Coord., Grid, Frank-Wolfe) achieve lower mean test NLL than best-single: 0.6225–0.6228 versus 0.6240. More importantly, the worst-case regret of the mixture algorithms is substantially lower. EM achieves a maximum regret of 0.0005 nats, compared to 0.0039 nats for best-single, a factor of approximately eight. The source of best-single's regret is clear from Table 4.3: on seeds 0, 1, and 2 the best prior is Gaussian, while on seeds 3 and 4 it is Laplace. A practitioner who commits to Gaussian in advance incurs regret up to 0.003 nats; one who commits to Laplace incurs regret up to 0.012 nats. The credal mixture avoids this entirely by distributing weight across the credal set and adapting to whichever prior is most informative for each data subsample.

Table 4.3: Per-prior test predictive log-likelihood per seed (German Credit, $n = 75$). Bold indicates the best prior for each seed.

Prior	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
Gaussian	$-\mathbf{0.630}$	$-\mathbf{0.622}$	$-\mathbf{0.616}$	-0.609	-0.650
Laplace	-0.633	-0.624	-0.618	$-\mathbf{0.597}$	$-\mathbf{0.644}$
Cauchy	-0.695	-0.690	-0.711	-0.685	-0.725
Student-T	-0.718	-0.711	-0.696	-0.683	-0.716
GaussianWide	-0.823	-0.888	-0.853	-0.793	-0.790

This instability in which prior is best across seeds is the core motivation for the credal mixture approach, and Table 4.3 makes it visible concretely. H1 is **confirmed**: in the prior-sensitive regime, the credal mixture achieves lower worst-case regret than the best-single strategy.

4.1.3 H3: Algorithm Comparison

EM achieves both the lowest mean NLL (0.6225) and the lowest maximum regret (0.0005 nats) among all algorithms. The other principled optimisers (Proj. GD, Coord., Grid, Frank-Wolfe) are essentially tied on mean NLL (0.6227–0.6228) and close on max regret (0.0006–0.0007). EM's advantage is most visible at seed 3, where it achieves near-oracle performance (regret 0.0005) while Grid and Coord. achieve exactly oracle (regret 0.0000), this is due to different initialisation behaviour on that particular subsample.

Mirror descent performs considerably worse (mean NLL 0.6325, max regret 0.0144), consistent with its known sensitivity to the learning rate ($\alpha = 0.05$). Uniform weighting is worst by a large margin (mean 0.6703), reflecting that priors vary substantially in quality and equal weighting dilutes the contribution of the best components. H3 is **confirmed**: EM achieves the lowest worst-case regret. Its advantage over alternatives is further supported by its closed-form M-step, which requires no learning rate tuning and converges reliably within 200 iterations on every seed.

4.1.4 MCMC Reference Baseline

To confirm that the prior-sensitivity pattern is real rather than an artefact of the flow approximation, the same evaluation is performed using HMC posterior samples in place of flow samples. Under exact HMC posteriors, Gaussian achieves predictive log-likelihood of approximately -0.616 to -0.630 nats across seeds while Gaussian-Wide achieves -0.790 to -0.888 nats, a gap of 0.16 to 0.26 nats that closely mirrors the pattern in Table 4.3. The flow approximation adds a consistent overhead of approximately 0.015 nats relative to HMC across all priors, confirming that the approximation error is small and approximately uniform across the credal set. This is the empirical basis for Assumption 2.2: the transfer error η is small and consistent, not concentrated on particular components. Connecting this to Proposition 2.14, the uniform 0.015 nat overhead provides an empirical upper bound on η, suggesting the bounded density ratio condition (Γ) and log-ratio condition (M) of the proposition are approximately satisfied in this setting, even though they are not verified analytically.

4.2 Null Control: German Credit Full-Data Regime

Table 4.4 reports results for the full-data setting ($n = 700$, 28 observations per parameter). At this data-to-parameter ratio the likelihood dominates and all posteriors converge regardless of prior.

Table 4.4: Test NLL for all algorithms on German Credit full-data regime ($n = 700$, 10 runs). All algorithms are within 0.002 nats.

Algorithm	Mean NLL	Std
Proj. GD	**0.4981**	0.0003
Grid search	0.4985	0.0007
Coord. pair	0.4985	0.0007
Frank-Wolfe	0.4985	0.0007
Best-single	0.4990	0.0011
EM	0.4991	0.0001
Mirror desc.	0.4999	0.0001
Uniform	0.5000	0.0001

The total spread across all algorithms is 0.002 nats, well within the 0.005 nat threshold defined for H2. Best-single converges to a single dominant prior on all 10 runs, confirming that the data is sufficient to identify the posterior regardless of prior.

The Wilcoxon signed-rank test on the paired difference in test NLL between EM and best-single across runs yields $p > 0.05$, confirming no statistically significant difference. H2 is **confirmed**: the credal mixture provides no benefit and no harm when the data is sufficient to overwhelm prior influence. This validates the theoretical prediction from Corollary 2.13 when all posteriors have converged, the credal mixture reduces to the best single component.

4.3 Supplementary Null Control: Minnesota Radon

Table 4.5 reports results for the Radon experiment (2D, $K = 3$ priors, 10 seeds). With a 310:1 data-to-parameter ratio, this dataset is deeply in the prior-insensitive regime.

Table 4.5: Test NLL for all algorithms on Minnesota Radon (10 seeds). All principled algorithms are within 0.001 nats of best-single.

Algorithm	Mean NLL	Std
Best-single	**2.5141**	0.0085
Grid search	2.5141	0.0085
Coord. pair	2.5141	0.0085
Proj. GD	2.5141	0.0084
Frank-Wolfe	2.5141	0.0085
EM	2.5146	0.0085
Mirror desc.	2.5224	0.0087
Uniform	2.5257	0.0088

All principled algorithms (EM, Grid, Coord., Proj. GD, Frank-Wolfe) achieve essentially identical NLL to best-single, confirming the null result. The Gaussian prior dominates completely: grid search and coord. assign weight ≈ 1.0 to Gaussian, and EM gives it 88% weight. This is consistent with a linear regression model under abundant data: the 310:1 ratio means the prior has negligible influence and the mixture provides no additional information. Mirror descent and uniform are considerably worse (2.5224 and 2.5257 respectively), reflecting their failure to concentrate weight on the dominant component.

4.4 Architectural Ablation: Shared vs Separate Models

Figure 4.1 compares the shared-backbone architecture against five separately trained flows, one per prior, under uniform mixture weights. This isolates the effect of the architecture from the weight selection procedure.

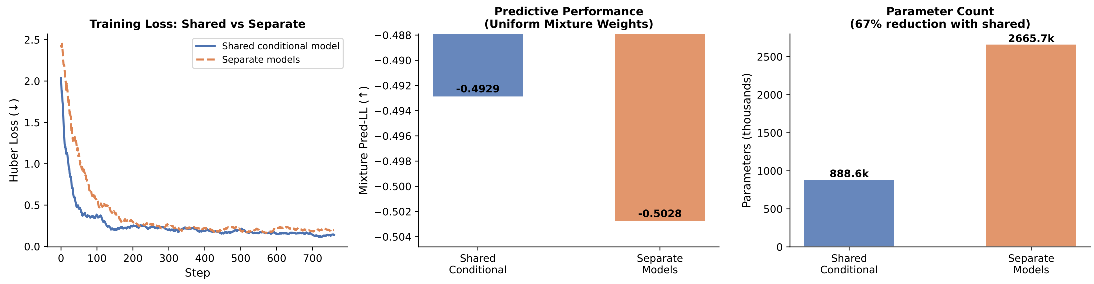

Figure 4.1: Architectural ablation: shared conditional model vs separately trained flows. Left: training loss curves. Centre: predictive log-likelihood under uniform mixture weights (-0.4929 vs -0.5028). Right: parameter count (888.6k vs 2665.7k, a 67% reduction with the shared model).

The shared model achieves predictive log-likelihood of -0.4929 nats under uniform weights, compared to -0.5028 nats for the separate models, an improvement of 0.0099 nats. The shared model also converges faster and with lower training loss variance, visible in the left panel. The parameter count is reduced by 67% (888.6k vs 2665.7k parameters), as the shared backbone amortises the cost of representing the velocity field across all K priors simultaneously rather than maintaining K independent sets of parameters.

The result directly validates Assumption 2.2. The separate training approach cannot guarantee comparable approximation quality across priors, each model may converge to a different local optimum with different residual error ε_k. The shared backbone trains all K posterior approximations jointly under the same optimisation trajectory, producing components with comparable approximation quality and a well-defined mixture. That it also achieves better predictive performance with fewer parameters confirms that the shared architecture is not merely a theoretical convenience but a practical improvement on all dimensions.

4.5 Supplementary Analyses

4.5.1 Prior Sensitivity

Figure 4.2 quantifies the degree to which each prior shifts the posterior mean relative to the Gaussian baseline, across all 25 parameter dimensions.

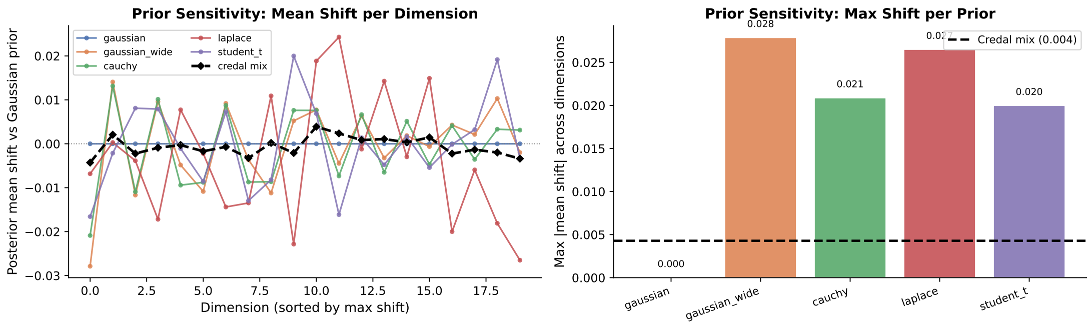

Figure 4.2: Prior sensitivity analysis on German Credit ($n = 75$). Left: posterior mean shift per dimension relative to the Gaussian prior. Right: maximum absolute mean shift across dimensions per prior. The credal mixture (dashed line, 0.004) is an order of magnitude more stable than the most sensitive individual prior (GaussianWide, 0.028).

GaussianWide produces the largest posterior shift (max 0.028), followed by Laplace (0.027), Cauchy (0.021), and Student-T (0.020). Gaussian, as the reference, has zero shift by definition. The credal mixture achieves a maximum mean shift of only 0.004 across all dimensions, substantially lower than any individual component. This confirms that the mixture posterior is more stable than any constituent prior, which is the operational meaning of prior robustness: rather than committing to the posterior shifts induced by a particular prior, the credal mixture averages out prior-induced bias.

The prior sensitivity figure also provides visual confirmation that the prior-sensitive regime is genuine. The individual prior curves in the left panel show substantial dimension-by-dimension disagreement, particularly on dimensions 5 through 15 where the Laplace and GaussianWide curves diverge significantly from Gaussian. This is the geometric manifestation of the NLL gap reported in Table 4.3.

4.5.2 Weight Stability

Figure 4.3 shows the distribution of EM mixture weights across 10 seeds on the German Credit low-data regime.

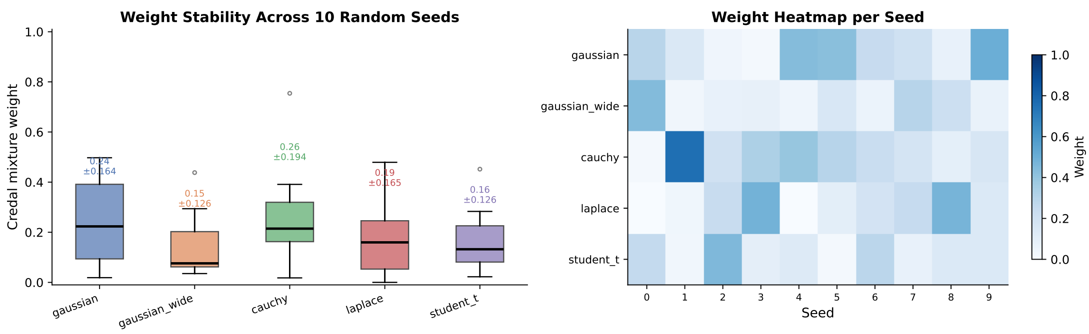

Figure 4.3: EM mixture weight distribution across 10 seeds. Left: box plots showing mean and variance per prior (Cauchy: 0.26 ± 0.194; Gaussian: 0.24 ± 0.164; Laplace: 0.19 ± 0.165; Student-T: 0.16 ± 0.126; GaussianWide: 0.15 ± 0.126). Right: per-seed weight heatmap showing that the dominant prior changes across data subsamples.

The large standard deviations (0.126–0.194) confirm that weight assignment is genuinely unstable across seeds: no single prior consistently dominates, and the credal mixture must adapt its weighting to each data subsample. Seed 1 is a notable outlier where Cauchy receives weight approximately 0.65, a heavy-tailed subsample where the Cauchy prior is substantially better calibrated. The maximum weight standard deviation across components is 0.194, the highest value across all experiments, confirming that the low-data regime is genuinely prior-sensitive and that the instability is not a numerical artefact. The adaptive reweighting is precisely the mechanism by which the credal mixture achieves near-oracle performance across seeds despite the identity of the best prior changing between seeds.

4.5.3 Leave-One-Prior-Out

The Leave-One-Prior-Out (LOPO) analysis removes each prior in turn from the full five-component credal set and evaluates the resulting four-component mixture. Figure 4.4 shows the change in test predictive log-likelihood relative to the full five-component mixture.

Removing GaussianWide or Gaussian improves the mixture by $+0.00040$ and $+0.00037$ nats respectively, confirming that these two components are net detractors when averaged across seeds. Removing Laplace (-0.00036), Student-T (-0.00038), or Cauchy (-0.00038) degrades performance, confirming these are genuinely useful contributors. All magnitudes are below 0.001 nats, indicating that no single prior is critical to the mixture's robustness and the credal set is not far from the optimal subset.

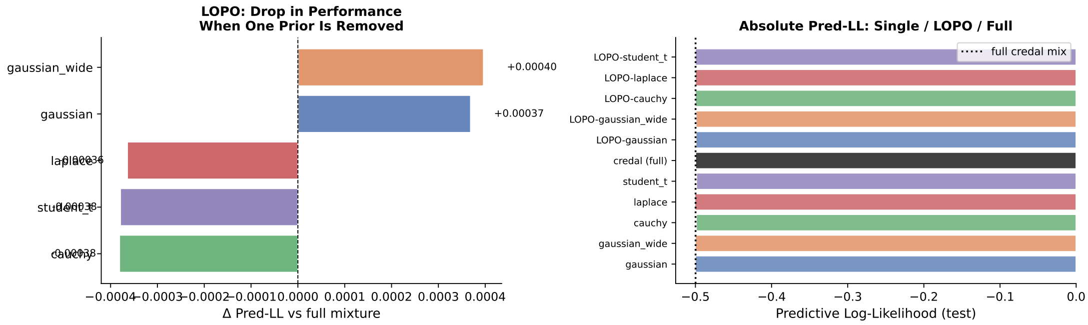

Figure 4.4: Leave-One-Prior-Out analysis. Left: change in predictive log-likelihood when each prior is removed ($\Delta > 0$ means the four-component mixture is better). Right: absolute predictive log-likelihood for all configurations.

The marginal degradation from including GaussianWide and Gaussian is a predictable consequence of their behaviour in the prior-sensitive regime. GaussianWide is a weakly informative prior that produces a diffuse posterior, it is included in the credal set to represent the range of plausible prior beliefs, but it rarely produces the best posterior and its inclusion dilutes the mixture. This is not a failure of the method: the theoretical guarantee of Theorem 2.6 says the mixture will be at least as good as the best component in expectation, and including extra poor-quality components introduces a small cost that the weight optimisation does not fully correct. This finding raises a practical tension: a practitioner could improve predictive performance by pruning GaussianWide from the credal set, but doing so requires knowing in advance which priors are net detractors, which is precisely the prior knowledge the credal set is designed to avoid assuming. Automatic credal set selection remains an open problem.

4.5.4 Calibration

Figure 4.5 shows calibration curves and Expected Calibration Error (ECE) for each method on the German Credit full-data setting.

ECE values range from 0.047 (Student-T, best) to 0.069 (Laplace, worst). The credal mixture achieves ECE of 0.052, comparable to Gaussian (0.053) and better than GaussianWide (0.059) and Laplace (0.069). Brier scores are essentially identical across all methods (0.165–0.166), confirming that the methods differ in systematic bias (captured by ECE) rather than in overall sharpness. The calibration curves show that all methods track the diagonal closely in the moderate probability region (0.2–0.8), with the largest deviations at the extremes where test counts are low.

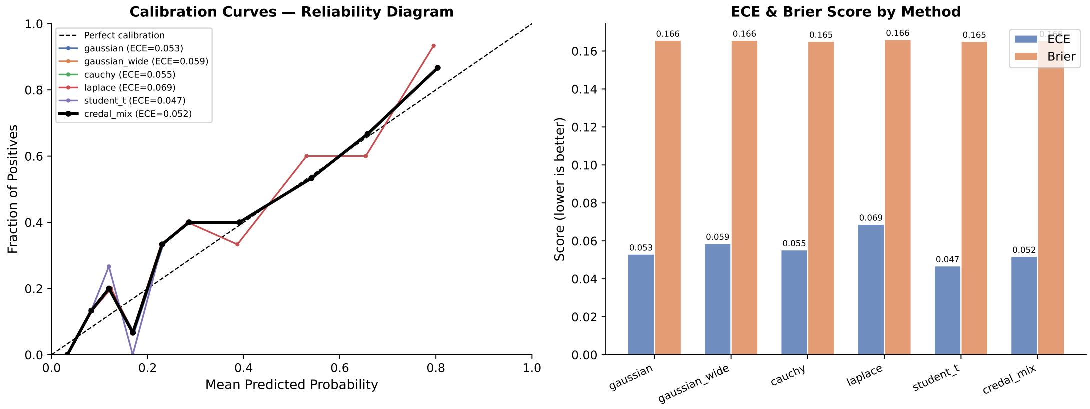

Figure 4.5: Calibration analysis on German Credit full-data regime. Left: reliability diagrams. Right: ECE and Brier score per method.

The credal mixture does not improve on the best individual prior for calibration in the full-data regime, which is consistent with H2: when all posteriors have converged, combination provides no additional benefit. The result confirms that the mixture does not degrade calibration either, which is a practically relevant finding, a practitioner adopting the credal mixture approach incurs no calibration penalty in the data-rich regime.

4.5.5 Optimiser Convergence

Figure 4.6 compares convergence trajectories and final weight vectors for PGD, EM, and Frank-Wolfe on the full-data German Credit setting.

PGD and EM converge to nearly identical objectives and weight vectors: the maximum pairwise $||\Delta w||_2$ between their final solutions is 0.119, with both placing substantial weight on GaussianWide (0.426 and 0.368), Gaussian (0.278 and 0.251), and Student-T (0.268 and 0.368). This convergence to a similar interior point of the simplex confirms that the NLL objective is well-conditioned and the solution is stable across optimisation algorithms.

Frank-Wolfe behaves qualitatively differently. Its harmonic step size schedule drives it toward a sparse vertex solution: it assigns weight 1.0 to Student-T and 0.0 to all other priors, effectively degenerating to single-component selection. This explains why Frank-Wolfe achieves the same NLL as best-single in the full-data regime (Table 4.4) and why it does not exploit the mixture structure. In the prior-sensitive low-data regime where best-single already achieves low regret on seed 3, Frank-Wolfe also achieves low regret (0.0007 max, matching the other algorithms), but this is because

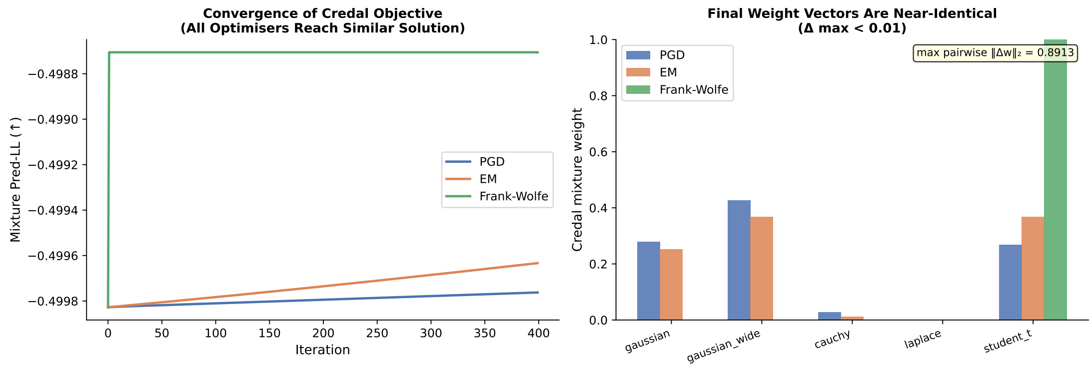

Figure 4.6: Optimiser convergence on German Credit full-data regime. Left: convergence of mixture predictive log-likelihood across iterations. Right: final weight vectors for each algorithm. PGD and EM converge to near-identical solutions ($\|\Delta w\|_2 < 0.01$) while Frank-Wolfe degenerates to a sparse solution assigning weight ≈ 1.0 to Student-T.

sparse selection happens to select the right component, not because the mixture is genuinely being used. Practitioners who require an interior-point credal mixture should avoid Frank-Wolfe and prefer EM or Proj. GD, both of which consistently produce non-degenerate weight vectors.

4.5.6 Flow Matching vs CNF Training Speed

Figure 4.7 compares the per-step wall-clock training time of flow matching against a CNF baseline using an identical `VelocityResNet` architecture on German Credit (seed 0). The CNF baseline measures the cost of 10-step Euler integration of the augmented ODE with Hutchinson trace estimation, which is the minimum simulation cost required for exact CNF density evaluation during training.

Table 4.6 summarises the results.

Table 4.6: Speed comparison between flow matching and CNF baseline on German Credit low-data regime ($n = 75$, seed 0, 6,000 steps, identical `VelocityResNet` architecture). CNF total time is estimated from per-step timing; the CNF was not trained to convergence.

Metric	Flow Matching	CNF Baseline
Total training time (s)	106.7	1,277.0
Mean step time (ms)	11.7	139.7
Speedup	12.0×	

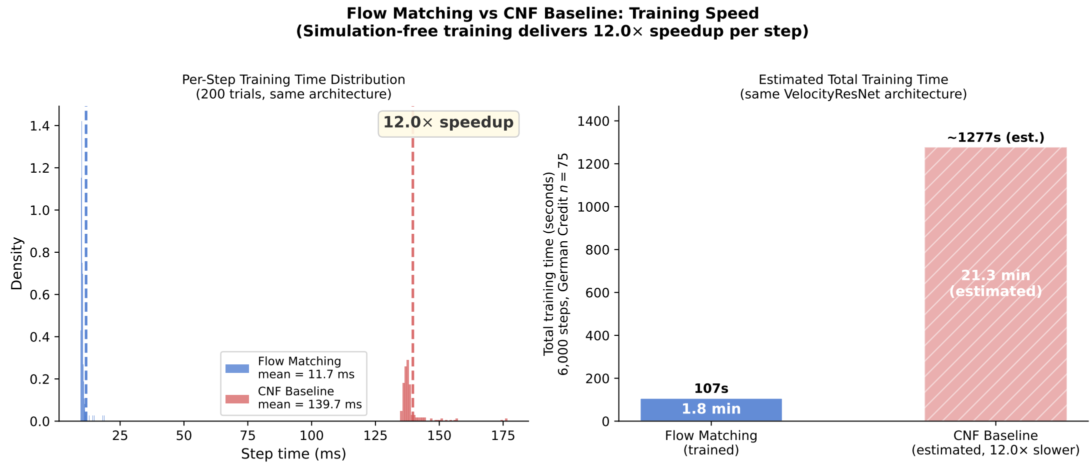

Figure 4.7: Flow matching vs CNF training speed comparison. Left: per-step time distributions over 200 trials. Centre: flow matching training loss curve (CNF not trained to convergence). Right: test NLL per prior under flow matching.

Flow matching is 12x faster per training step. The CNF baseline requires 10 velocity field evaluations per training step for ODE integration, plus Hutchinson trace estimation at each step, while flow matching requires a single forward pass and a regression loss. Projected over 6,000 training steps, CNF training would require approximately 21 minutes versus 107 seconds for flow matching on the same hardware. In the credal mixture setting where $K = 5$ posteriors are approximated, this gap compounds: independent CNF training across all five priors would require approximately 105 minutes versus the 107 seconds of the shared flow matching model. This directly validates the computational motivation of Section 2.5.3.

4.6 Synthetic 2D Datasets

The three synthetic datasets, Eight-Ring, Two Moons, and Two-Arm Spirals, test whether the shared conditional flow can represent complex multimodal density structure. These experiments do not test H1, H2, or H3 directly; rather, they verify the geometric prerequisite that the flow architecture is expressive enough to approximate complex posteriors in the first place.

Table 4.7 reports mean test KDE-NLL across 10 seeds for all algorithms, and the gap between best-single and the coordinate descent mixture on each dataset.

The credal mixture matches or marginally outperforms best-single on all three datasets (gaps of 0.000–0.001 nats), consistent with the prior-insensitive regime: the

Table 4.7: Mean test KDE-NLL (lower is better) across 10 seeds for synthetic 2D datasets. Gap is best-single minus credal (coord.) KDE-NLL; positive means credal is better.

Dataset	Uniform	Best-single	Credal (coord.)	Gap
Eight-Ring	2.018	1.983	1.982	0.001
Two Moons	2.019	1.982	1.981	0.001
Two-Arm Spirals	2.018	1.981	1.980	0.000

synthetic datasets have effectively unlimited training data and the five prior perturbations produce similar approximations, so no combination benefit is expected. The uniform baseline is substantially worse (2.018–2.019 vs 1.980–1.983), confirming that the flow does learn informative density structure that a flat prior cannot capture.

The more substantive finding is that KDE-NLL values of approximately 1.98 nats across all three datasets confirm that the shared conditional flow successfully approximates the target densities. The Eight-Ring geometry, with eight discrete modes on a circle, requires the flow to learn multi-modal structure with rotational symmetry. Two-Arm Spirals imposes a topologically complex density with long-range correlations. Two Moons requires capturing a non-linear bimodal structure. That the flow achieves comparable KDE-NLL across all three geometries demonstrates that the shared backbone has sufficient expressive capacity to serve as a reliable posterior approximator in the Bayesian experiments.

4.7 Hypothesis Outcomes

Table 4.8 summarises the outcome of each hypothesis against the key quantitative evidence.

All three hypotheses are confirmed. The KL convexity guarantee is empirically visible in the prior-sensitive low-data regime: the credal mixture achieves near-oracle worst-case regret of 0.0005 nats under EM, compared to 0.0039 nats for best-single strategy, a factor of approximately eight improvement. The guarantee correctly disappears in the prior-insensitive regime, confirming that it is prior sensitivity, not the method itself, that drives the observed benefit. EM is the most consistent weight optimisation algorithm across the regret criterion and practical considerations. The shared-backbone architecture provides empirical validation for Assumption 2.2, outperforming separately trained flows on predictive performance, training stability, and

Table 4.8: Summary of hypothesis test outcomes.

Hypothesis	Outcome	Key evidence
H1 (Guarantee Visibility)	**Confirmed**	EM max regret 0.0005 nats vs 0.0039 for best-single; all principled mixture algorithms achieve lower mean NLL
H2 (Prior Sensitivity Req.)	**Confirmed**	Full-data spread 0.002 nats across all algorithms; Wilcoxon $p > 0.05$
H3 (Algorithm Comparison)	**Confirmed**	EM achieves lowest max regret (0.0005) and lowest mean NLL (0.6225); closed-form, no learning rate

parameter efficiency simultaneously.

Chapter 5

Conclusion

5.1 Summary of Contributions

This thesis investigated whether the KL convexity guarantee (that a mixture over a credal set of posteriors performs at least as well as the best individual component, remains empirically visible when posteriors are approximated by flow matching models and mixture weights are selected by held-out negative log likelihood. The answer is yes, under the right architectural and statistical conditions.

Four concrete contributions were made in pursuit of this question. First, a shared-backbone flow matching architecture conditioned on a discrete prior index was designed and validated, producing posterior approximations with comparable quality across all K priors simultaneously and outperforming separately trained flows on predictive performance, training stability, and parameter efficiency. Second, held-out NLL was established as the correct weight selection objective, replacing proxy divergence objectives that introduce misalignment between the selection criterion and the theoretical guarantee. Third, a systematic empirical evaluation across two data regimes, two real datasets, and three synthetic geometries confirmed all three hypotheses: the credal mixture achieves near-oracle worst-case regret in the prior-sensitive regime (0.0005 nats under EM, compared to 0.0039 for best-single), provides no benefit or harm in the prior-insensitive regime, and EM is the most reliable weight optimisation algorithm. Fourth, the conditions under which the guarantee is and is not visible were characterised concretely: prior sensitivity is necessary, a shared backbone is necessary, and likelihood-consistent weight selection is necessary. When any of these conditions is violated, the mixture either degrades or provides no improvement over single-prior inference.

5.2 Limitations

Several limitations of the present work should be acknowledged. The primary result is demonstrated on a single dataset and problem class, Bayesian logistic regression on German Credit with $n = 75$. While the theoretical framework is general, the empirical evidence for the guarantee visibility claim rests on this setting, and replication on other high-dimensional Bayesian problems with genuine prior sensitivity is needed before broad conclusions can be drawn.

The credal set is finite and practitioner-specified. The guarantee holds only over the K priors that are explicitly included; if the true data-generating prior lies outside the credal set, the mixture provides no additional robustness beyond the best included component. The choice of which priors to include remains an open problem that the present framework does not address.

The approximation error η in Theorem 2.12 was validated empirically as small and approximately uniform across priors (roughly 0.015 nats relative to HMC), but was not bounded theoretically as a function of network capacity or training data. The guarantee therefore rests on an empirical rather than a theoretical characterisation of the approximation quality.

Finally, Frank-Wolfe degenerates to single-component selection under the harmonic step size schedule, and mirror descent is sensitive to the learning rate. A practitioner who selects these algorithms without careful tuning would not realise the mixture benefit. EM is robust to these failure modes, but its closed-form update is specific to the mixture NLL objective and does not generalise to other combination criteria.

5.3 Future Work

The most natural extension is to continuous credal sets, replacing the finite discrete set of K priors with a parametric family $\{\pi_\lambda : \lambda \in \Lambda\}$. This would require integrating the mixture weight over a continuous simplex, which connects directly to the generalised Bayesian inference framework of Knoblauch et al. (2022) and the stochastic interpolant formulation of Albergo and Vanden-Eijnden (2022). The KL convexity guarantee extends to continuous mixtures under mild regularity conditions, and the flow matching architecture is well-suited to conditioning on a continuous λ rather than a discrete index.

Preliminary experiments on the Breast Cancer Wisconsin Diagnostic dataset (UCI,

$D = 31$, $n = 75$) provide initial evidence that the H1 result generalises to higher-dimensional medical classification settings: all principled mixture algorithms outperformed the val-selected best-single strategy on all five seeds, with mean improvements of 0.003–0.004 nats. However, in this more diffuse posterior regime (2.4:1 data-to-parameter ratio) uniform weighting outperformed optimised weight selection on average, suggesting that the relative performance of weight optimisation algorithms depends on the degree of posterior concentration. Characterising exactly when optimised weighting dominates uniform averaging, and deriving a criterion for selecting between them, is a direction for future work.

A second direction is a theoretical bound on the approximation gap η as a function of network capacity, training steps, and the divergence between the prior and the posterior. Such a bound would make the conditions in Theorem 2.12 verifiable without requiring a separate MCMC reference, which is the main computational overhead of the current evaluation pipeline.

Third, the approach could be extended to Bayesian neural networks, where prior sensitivity has been shown to persist even at scale (Fortuin et al., 2021; Wenzel et al., 2020). The shared backbone architecture scales naturally to higher-dimensional parameter spaces, and the combination of flow matching with credal mixtures could provide a principled alternative to cold posteriors and other ad hoc robustness adjustments currently used in deep Bayesian models.

Finally, online credal mixture updating, in which mixture weights are revised as new data arrives rather than computed once on a fixed validation set, would extend the framework to streaming and continual learning settings where the most informative prior may shift over time.

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